6.2.4.5 Limitations. The fundamental current contributes to the useful torque of the machine, while the harmonics produce pulsating torques whose averages are zero and hence generate losses only. There will not be any triplen harmonics in the current in a star- (wye-) connected stator of the induction motor, but for the calculation of losses, thermal rating of the motor, and rating of the devices, the rms value of the current is required. It is computed approximately by using equation (6.5). Note that increasing triggering angles produce high losses in the machine that are due to the harmonics, thus affecting the safe thermal operation.

Calculation of the safe operating region of the machine requires a precise estimation of the rms current. In this regard, the approximate analysis is invalid for triggering angles greater than 135°. In such cases, it is necessary to resort to a rigorous analysis, using the dynamic equations of the motor instead of the steady-state equivalent circuit [2, 3, 4].

6.2.5 Torque–Speed Characteristics with Phase Control

Torque–speed characteristics as a function of triggering angle α, for a typical NEMA D, 75-hp, 4-pole, 460-V, 3-phase, 60-Hz machine are shown in Figure 6.8. The characteristics are necessary but not sufficient to determine the speed-control region available for a particular load. The thermal characteristics of the motor need to be incorporated to determine the feasible operating region. The thermal characteristics are dependent on the motor losses, which in turn are dependent on the stator currents. Adhering to the safe operating-current region enforces safe thermal operation. The stator current as a function of speed is derived later for use in application study. The parameters of the machine whose characteristics shown in Figure 6.8 are

$$R_s = 0.0862\ \Omega,\ R_r = 0.4\ \Omega,\ X_m = 11.3\ \Omega,\ X_{ls} = 0.295\ \Omega,\ X_{lr} = 0.49\ \Omega,$$
$$\text{full load slip} = 0.151, \text{and } I_b = 82.57\ \text{A}.$$

6.2.6 Interaction of the Load

The intersection of the load characteristics and torque–speed characteristic of the induction motor gives the operating point. A fan-load characteristic superimposed on the torque–speed characteristics of the induction motor whose parameters are given in Example 5.1 for various triggering angles with its upper value at 120° and lower value at 30° is shown in Figure 6.9. The feasible open-loop speed-control region is from 0.974 to 0.91 p.u., i.e., only 6.4% of the speed. To increase the speed-control region, either the operation has to be in the statically unstable region, or the load characteristics have to be modified. The latter is generally not possible. By using feedback control, the operating point can be anywhere, including the statically unstable region. Note that the speed is varied from 0.97 to 0.31 p.u. in the present case. The next step is to calculate the steady-state currents for a given load.

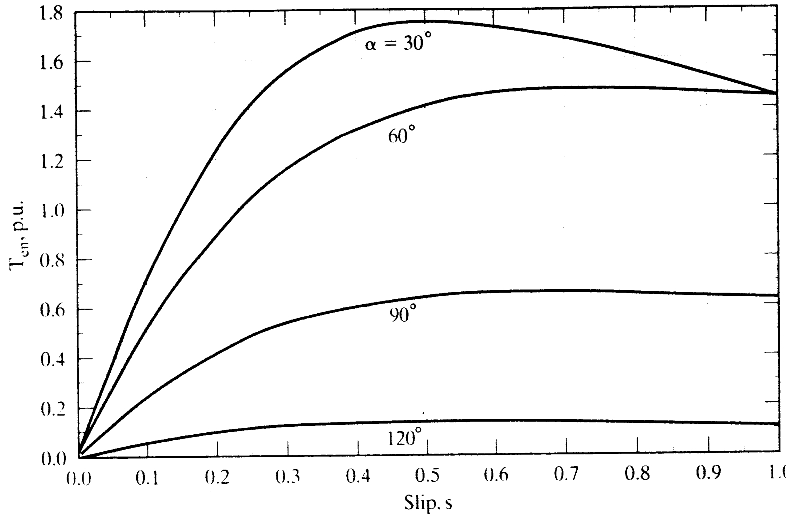


Figure 6.8 Normalized torque vs. slip as a function of triggering angle

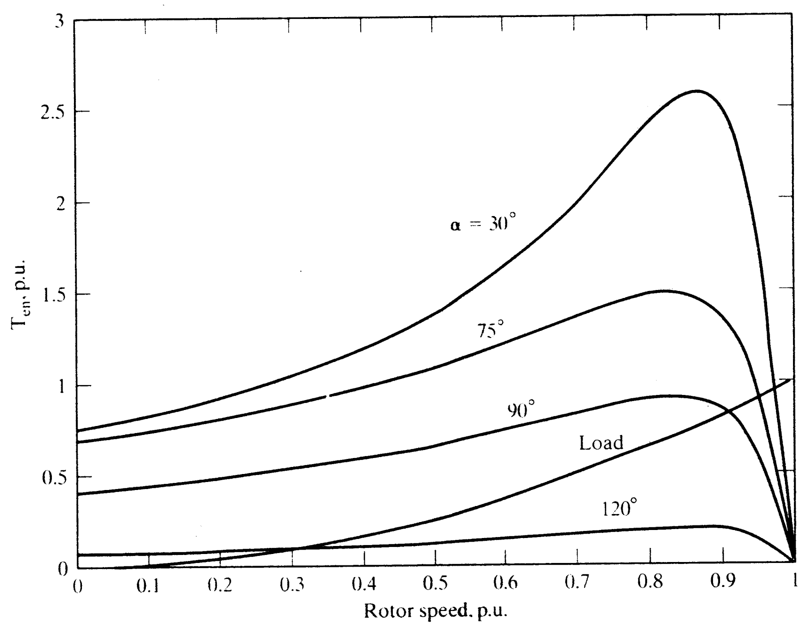


Figure 6.9 Torque–speed characteristics of the induction motor drive, and an arbitrary load to illustrate the speed-control region

6.2.6.1 Steady-state computation of the load interaction. Neglecting mechanical losses, the load torque equals the electromagnetic torque at steady state. The load torque, in general, is modeled as

$$T_l = B_l \omega_m^k \tag{6.26}$$

where the value of k is determined by the type of load, and B_l is the load constant. For instance,

$$\begin{aligned} k &= 0, \text{ for constant load torque} \\ &= 1, \text{ for friction load} \\ &= 2, \text{ for pump, fan loads} \end{aligned} \tag{6.27}$$

Equating the load torque to the electromagnetic torque and writing it in terms of the rotor current, speed, and motor constants gives

$$3\frac{PI_r^2R_r}{2s\omega_s} = B_l\omega_m^k \tag{6.28}$$

The rotor current in terms of the slip is derived from (6.28) as

$$I_r = \sqrt{\frac{B_l\omega_m^k s\omega_s}{3(P/2)R_r}} = K_r\sqrt{s(1-s)^k} \tag{6.29}$$

where

$$K_r = \sqrt{\omega_s^{k+1}\cdot\frac{B_l}{3(P/2)^{k+1}R_r}} \tag{6.30}$$

For a given value of k, the rotor current is maximum at a certain value of slip. It is found by differentiating equation (6.29) with respect to the slip and equating it to zero. The maximum currents and slip values for frictional and pump loads are

$$\begin{aligned} k &= 1, \quad I_{r(max)} = 0.50\ K_r, \text{ at } s = \frac{1}{2} \\ &= 2, \quad I_{r(max)} = 0.385\ K_r, \text{ at } s = \frac{1}{3} \end{aligned} \tag{6.31}$$

The stator current is computed from the rotor-current magnitude obtained from equation (6.29) by using the induction-machine equivalent circuit. Note that the air gap voltage is the product of the rotor current and rotor impedance. The magnetizing current is found from the air gap voltage and magnetizing impedance. The stator current is realized as the phasor sum of the magnetizing and rotor currents, leading to the following expression:

$$I_{as} = I_r\left[\frac{R_r}{s} + jX_{lr}\right]\left[\frac{1}{\frac{R_r}{s} + jX_{lr}} + \frac{1}{jX_m}\right] \tag{6.32}$$

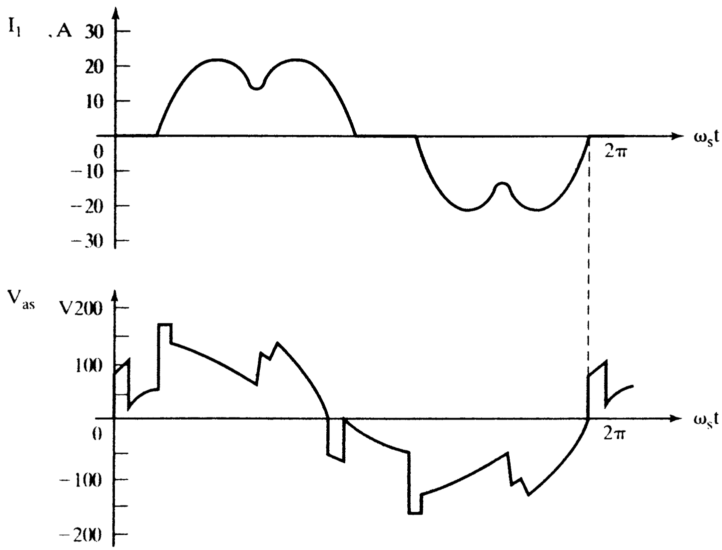


Figure 6.10 Typical line-current and phase-voltage waveforms of a phase-controlled induction motor drive

Typical steady-state phase-voltage and line-current waveforms are shown in Figure 6.10. These waveforms portray the actual condition of the motor drive, whereas the preceding analysis considered only the fundamental component.

Example 6.1

A speed-control range from 0.45 to 0.8 times full-load speed is desired for a pump drive system. A NEMA D induction motor is chosen with the following parameters:

$$150 \text{ hp}, 460 \text{ V}, 3 \text{ ph}, 60 \text{ Hz}, 4 \text{ poles, star connected}, R_s = 0.03\ \Omega, R_r = 0.22\ \Omega,$$
$$X_m = 10.0\ \Omega, X_{ls} = 0.1\ \Omega, X_{lr} = 0.12\ \Omega, \text{full load slip} = 0.1477, \text{friction and}$$
$$\text{windage losses} = (0.01\omega_m + 0.0005\omega_m^2)\ (\text{N}\cdot\text{m})$$

The pump load constant, B_l, is 0.027 N·m/(rad/sec)^2. Find the range of triggering angle required to achieve the desired speed variation with stator phase control.

Solution This problem is the inverse of the procedure described in the text, which outlines a method to find the torque–speed characteristics for various triggering angles and to determine the speed range available by superposing the load torque–speed characteristics on the motor characteristics. In this problem, from the load torque–speed characteristics, the determination of triggering angle range is attempted as follows.

The upper speed is 0.8 p.u. which should correspond to a triggering angle of α_1; the lower speed is 0.45 p.u., corresponding to a triggering angle of α_2. The rated speed is given by

$$\omega_m = \omega_s(1 - s) \times \frac{2}{P} = (2\pi \times 60)(1 - 0.1477) \times \frac{2}{4} = 160.6 \text{ rad/s}$$

Torque corresponding to 0.8 p.u. speed ($\omega_{m1} = 128.5$ rad/sec) is given by

$$T_{l1} = B_l\omega_{m1}^2 + 0.01\omega_{m1} + 0.0005\omega_{m1}^2 = 455.5 \text{ N}\cdot\text{m}.$$

Similarly, torque corresponding to 0.45 p.u. speed (ω_{m2} = 72.3 rad/sec) is

$$T_{l2} = B_l\omega_{m2}^2 + 0.01\omega_{m2} + 0.0005\omega_{m2}^2 = 144.35 \text{ N} \cdot \text{m}.$$

From the electromagnetic torques, the rotor and stator currents, stator phase angles, and phase voltages are calculated:

T_e, N·m	I_r, A	I_{as}, A	ϕ, deg	V_{as}, V
T_{l1} = 455.5	201.3	204.2	20.6	153.2
T_{l2} = 144.35	157.6	159.6	31.4	70.7

The triggering angles of α_1 and α_2 with respective conduction angles β_1 and β_2 yield the two stator voltages for chosen operating conditions. There are four unknowns ($\alpha_1, \alpha_2, \beta_1, \beta_2$) and two known values of v_{as}, so it is necessary to resort to Table 6.1 to find various choices of α and β for the given phase angles. They then are substituted in the voltage equation to verify that they yield the desired voltages. By that procedure,

$$\alpha_1 \cong 102° \text{ with } \beta_1 \cong 98°$$

$$\alpha_2 \cong 135° \text{ with } \beta_2 \cong 68°$$

are obtained, thus giving the range of triggering angle variation as 102° to 135°.

Example 6.2

Determine the current-vs.-slip characteristics for the phase-controlled induction motor drive whose details are given in Example 6.1. Assume fan and frictional loads, and evaluate the load constant based on the fact that 0.2 p.u. torque is developed at 0.7 p.u. speed.

Solution

$$\text{Base voltage, } V_b = \frac{V_{ll}}{\sqrt{3}} = \frac{460}{\sqrt{3}} = 265.58 \text{ V}$$

$$\text{Base current, } I_b = \frac{P_b}{3V_b} = \frac{150 \times 745.6}{3 \times 265.58} = 140.37 \text{ A}$$

$$\text{Base speed, } \omega_b = \frac{2\pi \times 60}{P/2} = \frac{2\pi \times 60}{4/2} = 188.49 \text{ rad/sec}$$

$$\text{Base torque, } T_b = \frac{P_b}{\omega_b} = \frac{150 \times 745.6}{188.49} = 593.33 \text{ N·m.}$$

Case (i): Friction load

$$\text{Load constant, } B_l = \frac{0.2T_b}{0.7\omega_b} = \frac{0.2 \times 593.33}{0.7 \times 188.49} = 0.8994 \text{ N·m/(rad/sec)}$$

$$K_r = \sqrt{\omega_s^2 \frac{B_l}{3\left(\frac{P}{2}\right)^2 \cdot R_r}} = 220.03$$

$$I_{rn} = \frac{K_r\sqrt{s(1-s)}}{I_b} = 1.5675\sqrt{s(1-s)} \text{ p.u.}$$

The normalized stator current is

$$I_{asn} = I_{rn}\sqrt{\left(\frac{R_r}{s}\right)^2 + X_{lr}^2}\left[\frac{1}{\frac{R_r}{s} + jX_{lr}} + \frac{1}{jX_m}\right] \text{ (p.u.)}$$

Case (ii): Fan load

$$\text{Load constant, } B_l = \frac{0.2T_b}{(0.7\omega_b)^2} = 0.0068 \text{ Nm/(rad/sec)}^2$$

$$K_r = \sqrt{\omega_s^3 \frac{B_l}{3\left(\frac{P}{2}\right)^3 \cdot R_r}} = 262.99$$

$$I_{rn} = \frac{K_r\sqrt{s(1-s)^2}}{I_b} = 1.8735\sqrt{s(1-s)} \text{ p.u.}$$

The stator current is computed as in the friction-load case. The normalized current-vs.-slip characteristics are shown in Figure 6.11. Note that, at synchronous speed, the current drawn is the magnetizing current: the rotor current is zero. It is to be noted that the speed-control range is very limited in phase-controlled drives for currents lower than the rated value.

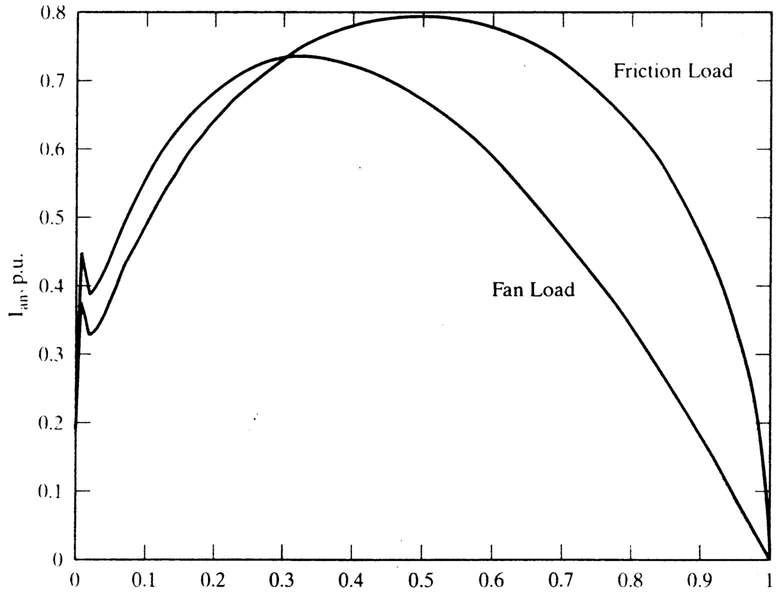


Figure 6.11 Normalized fundamental stator current vs. slip for friction and fan loads

6.2.7 Closed-Loop Operation

Feedback control of speed and current, shown in Figure 6.12, is employed to regulate the speed and to maintain the current within safe limits. The inner current-feedback loop is for the purpose of current limiting. The outer speed loop enforces the desired speed in the motor drive. The speed command is processed through a soft start/stop controller to limit the acceleration and deceleration of the drive system. The speed error is processed, usually through a PI-type controller, and the resulting torque command is limited and transformed into a stator-current command. Note that for zero torque command, the current command is not zero but equals the magnetizing current. To reduce the no-load running losses, it is advisable to run at low speed and at reduced flux level. This results in lower running costs compared to rated speed and flux operation at no load.

The current command is compared with the actual current, and its error is processed through a limiter. This limiter ensures that the control signal v_c to the phase controller is constrained to a safe level. The rise and fall of the control-signal voltage are made gradual, so as to protect the motor and the phase controller from transients. In case there is no feedback control of speed, the control voltage is increased at a preset rate. The current limit might or might not be incorporated in such open-loop drives.

6.2.8 Efficiency

Regardless of the open-loop or closed-loop operation of the phase-controlled induction motor drives, the efficiency of the motor drive is proportional to speed. The efficiency is derived from the steady-state equivalent circuit of the induction

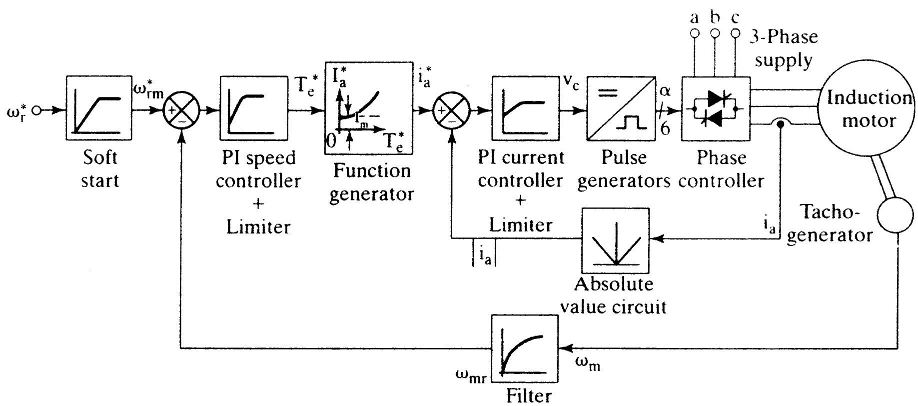


Figure 6.12 Closed-loop schematic of the phase-controlled induction motor drive

motor. Neglecting stator copper losses, stray losses, friction, and windage losses, the maximum efficiency is given as

$$\text{Efficiency, } \eta = \frac{P_s}{P_a} = \frac{P_m - P_{fw}}{P_a} = \frac{I_{rn}^2 \cdot \dfrac{(1-s)}{s} \cdot R_{rn}}{\dfrac{I_{rn}^2}{s} \cdot R_{rn}} = (1 - s) = \omega_{rn} \quad (6.33)$$

Here, P_a and P_m denote the air gap and the mechanical output power of the induction motor, respectively, and P_{fw} denotes the friction and windage losses. Slip variation means speed variation, and it is seen from (6.33) that the efficiency decreases as the speed decreases. The rotor copper losses are equal to slip times the input air gap power. This imposes severe strain on the thermal capability of the motor at low operating speeds. Operation over a wide speed range will be restricted by this consideration alone more than by any other factor in this type of motor drive.

The efficiency can be calculated more precisely than is given by equation (6.33) in the following manner:

$$\eta = \frac{P_m - P_{fw}}{P_m + P_{rc} + P_{sc} + P_{co} + P_{st}} \quad (6.34)$$

where

$$\left.\begin{aligned} P_m &= \text{Power output} = \frac{3I_r^2R_r(1-s)}{s} \\ P_{rc} &= \text{Rotor copper losses} = 3I_r^2R_r \\ P_{sc} &= \text{Stator copper losses} = 3I_{as}^2R_s \\ P_{co} &= \text{Core losses} = 3I_c^2R_c \end{aligned}\right\}, \quad (6.35)$$

P_{st} is the stray losses, and the shaft output power is the difference between the mechanical power and the friction and windage losses.

This calculation yields a realistic estimate of efficiency. Energy savings are realized by reducing the motor input compared to the conventional rotor-resistance or stator-resistance control. Consider the case where speed is controlled by adding an adjustable external resistor in the stator phases. That, in turn, reduces the applied voltage to the motor windings. Considerable copper losses occur in the external resistor, R_{ex}. In the phase-controlled induction motor, the input voltage is varied without the accompanying losses, as in the case of the external resistor based speed control system. The equivalent circuit of the resistance-controlled induction motor is shown in Figure 6.13.

The equivalent normalized resistance and reactance of the induction motor are obtained from equations (6.2) and (6.3). The applied voltage is then written, from the equivalent circuit shown in Figure 6.13, as

$$V_{as} = I_{as}(R_{ex} + R_{im} + jX_{im}) \quad (6.36)$$

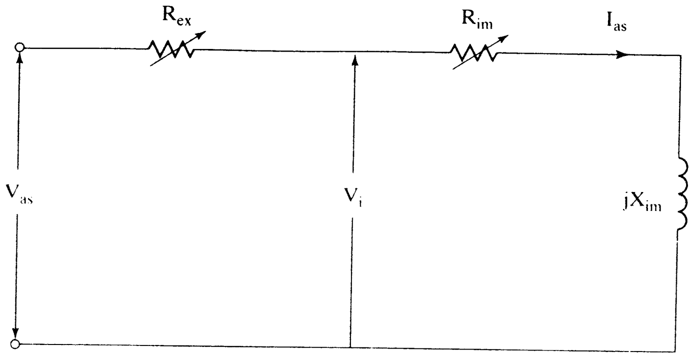


Figure 6.13 Equivalent circuit of the stator-resistance-controlled induction motor

The external resistor value for each slip is found by substituting for rated applied voltage and the stator current required to meet the load characteristics. The latter is obtained from the rotor current, calculated from load constants and slip given by (6.28) and (6.29). Then the stator current is computed from equation (6.32) for each value of slip. Summarizing these steps gives the following:

$$I_r(s) = K_r\sqrt{s(1 - s)^k} \tag{6.37}$$

$$I_{as}(s) = \frac{\sqrt{\left(\frac{R_r}{s}\right)^2 + X_r^2}}{X_m} \cdot I_r(s) \tag{6.38}$$

The external resistor for each operating slip is

$$R_{ex} = \sqrt{\left(\frac{V_{as}}{I_{as}(s)}\right)^2 - X_{im}^2} - R_{im} \tag{6.39}$$

The energy savings by using phase control of the stator voltages are approximately

$$P_{ex} = 3I_{as}^2R_{ex} \tag{6.40}$$

where I_{as} is obtained from equation (6.38) and R_{ex} is obtained from equation (6.40).

For Example 6.1, consider a fan load requiring 0.5 p.u. torque at rated speed. The speed variation possible with external stator-resistor control is 0 to 0.35 p.u. For this range of speed variation, the external resistor required and normalized power input, P_{in}, and power savings, P_{exn}, are shown in Figure 6.14(i). The efficiency of the drive with stator-phase and resistor control is shown in Figure 6.14(ii). The efficiency of the stator-phase control is computed by subtracting the external resistor losses from the input and treating that as the input in the stator-phase control case. The fact that stator-phase control is very much superior to the stator-resistor control is clearly demonstrated from Figure 6.14(ii).

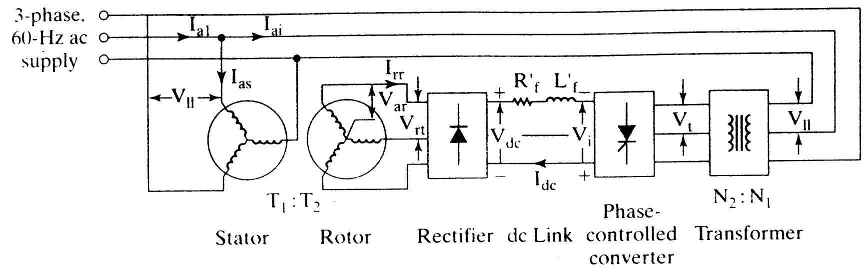


Figure 6.16 Slip-power recovery scheme

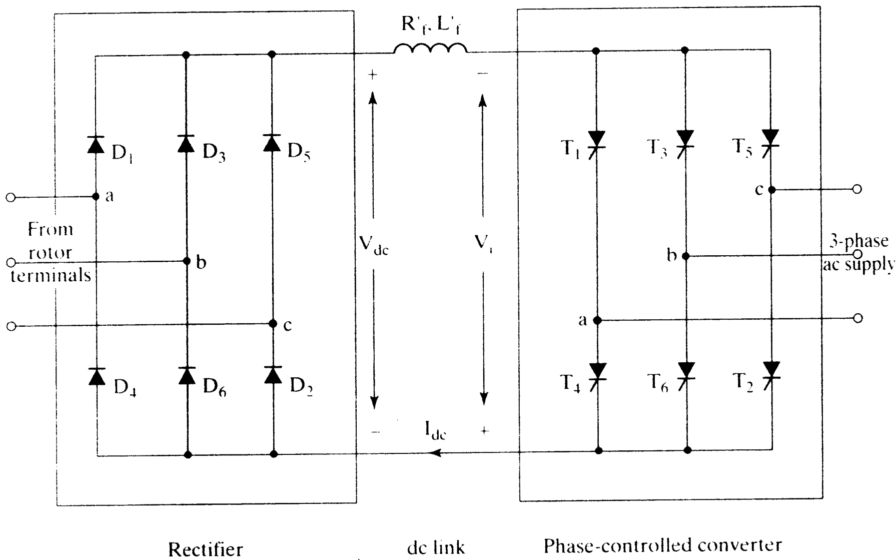


Figure 6.17 Rectifier–inverter power stage for slip-power recovery scheme

terminals. The rotor voltages are rectified and, through an inductor, fed to the phase-controlled converter. The detailed diagram of the bridge-diode rectifier and phase-controlled converter are shown in Figure 6.17. The operation of the phase-controlled converter has been described in Chapter 3.

The phase-controlled converter is operated in the inversion mode by having a triggering angle greater than 90°. Then the inverter voltage V_i is in opposition to the rectified rotor voltage V_{dc} in the dc link. A current I_{dc} is established in the dc link by adjusting the triggering angle of the phase-controlled converter. The diode-bridge rectifier on the rotor side of the induction motor forces power to flow only away from the rotor windings. A transformer is generally introduced between the phase-controlled converter and the ac supply to compensate for the low turns ratio in the induction motor and to improve the overall power factor of the system. The latter assertion will be proved in the subsequent section.

6.3.3 Steady-State Analysis

The equivalent circuit is used to predict the steady-state motor drive performance. The steady-state performance is computed by proceeding logically from the rotor part of the induction motor to the stator part through the converter subsystem. The dc link current is assumed to be ripple-free.

The rotor phase voltage, V_{ar}, and the dc link current, I_{dc}, are shown in Figure 6.18(a). The current is a rectangular pulse of 120-degree duration. The magnitude of the rotor current is equal to the dc link current, I_{dc}. The phase and line voltages and phase current in the phase-controlled converter are shown in Figure 6.18(b). There is a phase displacement of α degrees between the phase voltage and current, which is the triggering angle in the converter. The magnitude of the converter line current is equal to the dc link current but has 120-degree duration for each half-cycle. The rms rotor line voltage in terms of stator line voltage is

$$V_{rt} = \left(\frac{k_{w2}T_2}{k_{w1}T_1}\right)sV_{ll} \tag{6.42}$$

where k_{w1} are k_{w2} are the stator and rotor winding factors and T_1 and T_2 are the stator and rotor turns per phase, respectively.

Denoting the effective turns ratio by a, we write

$$\frac{1}{a} = \left(\frac{k_{w2}T_2}{k_{w1}T_1}\right) \tag{6.43}$$

The rotor line voltage is then derived from equations (6.42) and (6.43) as

$$V_{rt} = \frac{sV_{ll}}{a} \tag{6.44}$$

The average rectified rotor voltage is

$$V_{dc} = 1.35V_{rt} = \frac{1.35sV_{ll}}{a} \tag{6.45}$$

This has to be equal to the sum of the inverter voltage and the resistive drop in the dc link inductor.

Neglecting the resistive voltage drop, we get

$$V_{dc} = -V_i \tag{6.46}$$

where the phase-controlled-converter output voltage is given by

$$V_i = 1.35V_t = \cos\alpha \tag{6.47}$$

where

$$V_t = \left(\frac{N_2}{N_1}\right)V_{ll} = n_tV_{ll} \tag{6.48}$$

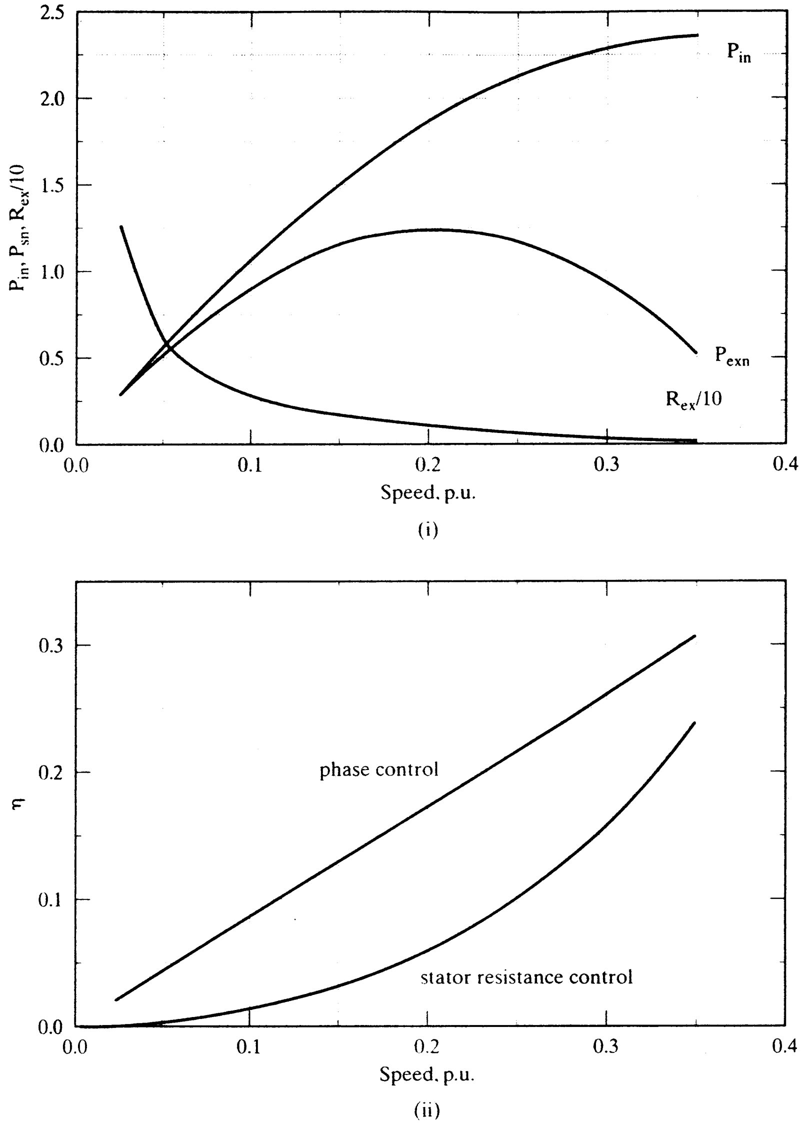


Figure 6.14 **(i)** External resistor, power input, and power savings vs. slip characteristics at rated voltage input **(ii)**Efficiency with external resistor and phase control vs. slip

6.2.9 Applications

Current-limited phase-controlled induction motor drives find applications in pumps, fans, compressors, extruders, and presses. The closed-loop drives are used in ski-lifters, conveyors, and wire-drawing machines. The phase controllers are used as solid-state contactors and also as starters in induction motors. Phase controllers are extensively used in resistance heaters, light dimmers, and solid-state relaying.

For obtaining maximum speed-control range in open-loop operation, NEMA D induction motors are used. These motors have a high rotor resistance and hence provide a high starting torque as well as a very large statically stable operating torque–speed region, but these motors have higher losses than other motors and hence have poorer efficiency.

6.3 SLIP-ENERGY RECOVERY SCHEME

6.3.1 Principle of Operation

The phase-controlled induction motor drive has a low efficiency; its approximate maximum efficiency equals the p.u. speed. As speed decreases, the rotor copper losses increase, thus reducing the output and efficiency. The rotor copper losses are

$$P_{rc} = 3I_r^2R_r = sP_a \tag{6.41}$$

where P_a is the air gap power. The increase in this slip power, sP_a, results in a large rotor current. This slip power can be recovered by introducing a variable emf source in the rotor of the induction motor and absorbing the slip power into it. By linking the emf source to ac supply lines through a suitable power converter, the slip energy is sent back to the ac supply. This is illustrated in Figure 6.15. By varying the magnitude of the emf source in the rotor, the rotor current, torque, and slip are controlled. The rotor current is controlled and hence the rotor copper losses, and a significant portion of the power that would have been dissipated in the rotor is absorbed by the emf source, thereby improving the efficiency of the motor drive.

6.3.2 Slip-Energy Recovery Scheme

A schematic of the slip-energy recovery in the induction motor is shown in Figure 6.16. The rotor has slip rings, and a diode-bridge rectifier is connected to the slip-ring

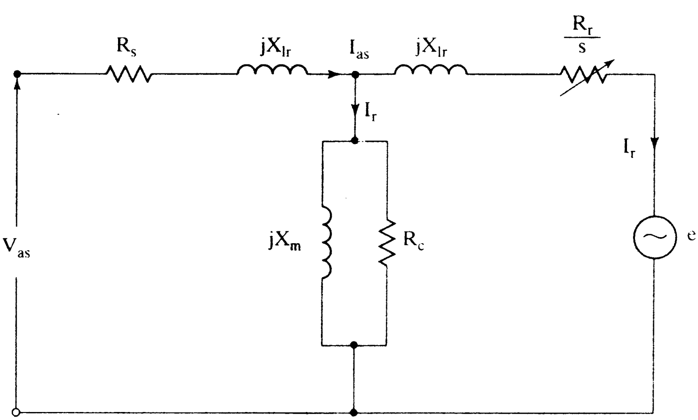


Figure 6.15 Steady-state equivalent circuit of the induction motor with an external induced-emf source in the rotor

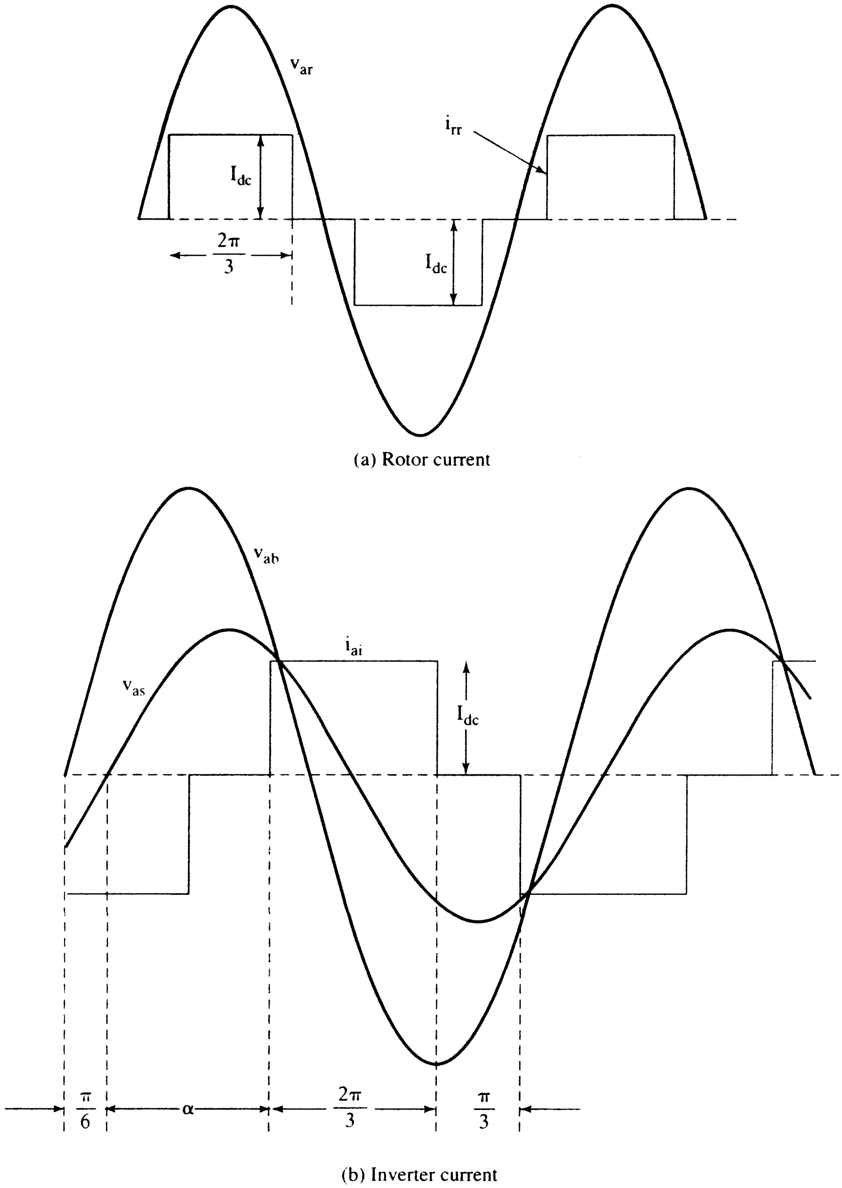


Figure 6.18 Rotor current and inverter current

and n_t is the turns ratio of the transformer. From equations (6.45) to (6.48), the slip is derived as

$$s = -(an_t)\cos\alpha \tag{6.49}$$

6.3.3.1 Range of slip. The triggering angle can be varied in the inversion mode from 90 to 180 degrees. The finite turn-off time of the power switches in the phase-controlled converter usually limits the triggering angle to 155 degrees. Hence, the practical range of the triggering angle is

$$90^\circ \leq \alpha \leq 155^\circ \tag{6.50}$$

For this range of α, the slip range is given by

$$0 \leq s \leq 0.906(an_t) \tag{6.51}$$

The turns ratio of the stator to rotor of the induction motor is usually less than unity. The dc link voltage becomes small when the induction motor is operating at around synchronous speed. This necessitates a very small range of triggering angle in the converter operation. A step-down transformer with a low turns ratio ensures a wide range of triggering angle under this circumstance.

The value of α in terms of the slip from equation (6.49) is

$$\alpha = \cos^{-1}\left\{-\frac{s}{an_t}\right\} \tag{6.52}$$

6.3.3.2 Equivalent circuit. The equivalent circuit for the slip-energy recovery scheme is derived by converting the dc link filter and phase-controlled inverter into their three-phase equivalents in steady state. The dc filter inductor is not required to be incorporated, as its effect in steady state is zero. The dc link filter resistance R_f' is converted to an equivalent three-phase resistance R_{ff} by equating their losses as follows:

$$I_{dc}^2 R_f' = 3(I_{rr})^2 R_{ff} \tag{6.53}$$

where I_{rr} is the rms value of the induction-motor rotor-phase current, which, in terms of the dc link current, is expressed as

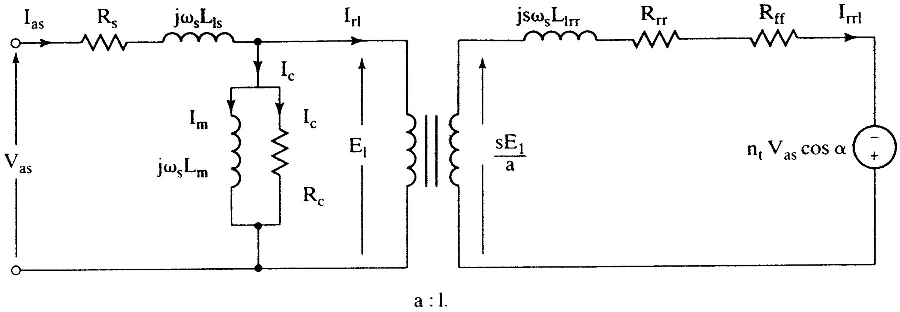
$$I_{rr} = \sqrt{\frac{1}{\pi}\int_0^{2\pi/3} I_{dc}^2\, d\theta} = I_{dc}\sqrt{\frac{2}{3}} = 0.816 I_{dc} \tag{6.54}$$

Substituting (6.54) into (6.53) gives R_{ff} as

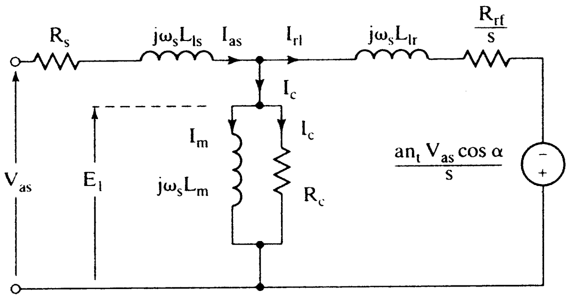
$$R_{ff} = 0.5R_f'$$

The rms value of the fundamental component of the rectangular rotor current is given by

$$I_{rr1} = \frac{4}{\pi\sqrt{2}} I_{dc}\sin 60^\circ = 0.779 I_{dc} \tag{6.55}$$



(i)



(ii)

Figure 6.19 Per-phase equivalent circuit of the slip-energy-recovery-controlled induction motor drive

The power transferred through the phase-controlled inverter is

$$P_{fb} = V_i I_{dc} = (1.35 n_t V_s \cos\alpha) I_{dc} = [1.35 n_t \sqrt{3} V_{as} \cos\alpha]\left[\frac{I_{rrl}}{0.779}\right] = 3[n_t V_{as} \cos\alpha] I_{rrl} \quad (6.56)$$

This could be viewed as power transfer to a three-phase ac battery with a phase voltage of $[n_t V_{as} \cos\alpha]$ absorbing a charging current of I_{rrl} from the rotor phases of the induction motor. By integrating R_{ff} and the emf of the variable ac battery in the three-phase rotor of the induction motor, the per-phase equivalent circuit shown in Figure 6.19(i) is obtained.

The reactive power demand of the variable emf source is met not from the stator and rotor of the induction motor but through the ac supply lines. Hence, that need not appear in the equivalent circuit of the slip-energy-recovery-controlled induction motor drive. Further, note that the rotor current is in phase with the variable emf source given by $[n_t V_{as} \cos\alpha]$. The equivalent circuit is further simplified and referred to the stator in a similar manner, as described in Chapter 5, with the following steps, as shown in Figure 6.19 (ii):

$$[I_{rrl}(R_{rr} + R_{ff}) - n_t V_{as} \cos\alpha] + js\omega_s L_{lrr} I_{rrl} = \frac{sE_1}{a} \quad (6.57)$$

The emf term is combined with the resistive voltage drop because it is in phase with I_{rr1}: the voltage component of the battery has only a real part in it, and the reactive part has been removed. This is an important point in the solution of rotor current from the derived equivalent circuit.

Multiplying (6.57) by $\frac{a}{s}$, we get

$$\left[I_{rr1}\left\{\frac{a(R_{rr}+R_{ff})}{s}\right\}-\frac{an_tV_{as}\cos\alpha}{s}\right]+j\omega_s aL_{lrr}I_{rr1}=E_1 \tag{6.58}$$

The rotor current, referred to the stator, is given by

$$I_{r1}=\frac{I_{rr1}}{a} \tag{6.59}$$

The stator-referred rotor and filter resistances and rotor leakage inductance are given by

$$\begin{aligned}R_r&=a^2R_{rr}\\R_f&=a^2R_{ff}\\L_{lr}&=a^2L_{lrr}\end{aligned} \tag{6.60}$$

and let

$$R_{rf}=R_r+R_f \tag{6.61}$$

Combining (6.59), (6.60), and (6.61) with (6.58) yields

$$\frac{I_{r1}}{s}\{R_{rf}-an_tV_{as}\cos\alpha\}+j\omega_s L_{lr}I_{r1}=E_1 \tag{6.62}$$

6.3.3.3 Performance characteristics. The performance characteristics can be evaluated from the equivalent circuit derived earlier. The electromagnetic torque is evaluated from the output power and rotor speed as

$$\begin{aligned}T_e=\frac{P_m}{\omega_m}=\frac{P_a(1-s)}{\left\{\dfrac{\omega_s(1-s)}{P/2}\right\}}=\frac{P}{2}\cdot\frac{P_a}{\omega_s}=\frac{P}{2}\cdot\frac{1}{\omega_s}\left\{3\left[I_{r1}^2\cdot\frac{R_{rf}}{s}-\frac{aI_{r1}n_tV_{as}\cos\alpha}{s}\right]\right\}\\=\frac{3P}{2}\cdot\frac{1}{s\omega_s}[I_{r1}^2R_{rf}-aI_{r1}n_tV_{as}\cos\alpha],\ \mathrm{N\cdot m}\end{aligned} \tag{6.63}$$

where P_a is the air gap power, P_m is the mechanical output power, ω_m is the mechanical rotor speed, ω_s is the synchronous speed, and P is the number of poles. By neglecting the leakage reactance of the rotor, the electromagnetic torque can be written as

$$T_e=\frac{3P}{2}\cdot\frac{1}{\omega_s}\cdot I_{r1}\cdot\left[\frac{I_{r1}R_{rf}}{s}-\frac{an_tV_{as}\cos\alpha}{s}\right]\cong\frac{3P}{2}\cdot\frac{1}{\omega_s}\cdot I_{r1}\cdot E_1 \tag{6.64}$$

If stator impedance is neglected, then the applied voltage V_{as} is equal to the induced emf E_1; hence,

$$T_e \cong \frac{3P}{2} \cdot \frac{V_{as}}{\omega_s} \cdot I_{r1} \tag{6.65}$$

and

$$I_{r1} = \frac{I_{rr1}}{a} = \frac{0.779 I_{dc}}{a} \tag{6.66}$$

which, on substitution, gives the electromagnetic torque as

$$T_e \cong \left(\frac{1.17P}{a}\right)\left(\frac{V_{as}}{\omega_s}\right) I_{dc} = K_t I_{dc} \tag{6.67}$$

where K_t is a torque constant given by

$$K_t = \left(\frac{1.17P}{a}\right)\left(\frac{V_{as}}{\omega_s}\right) \text{ (N·m/A)} \tag{6.68}$$

This shows that, for a given induction motor, the electromagnetic torque is proportional to the dc link current. An implication of this result is that torque control in this drive scheme is very similar to the torque control in a separately-excited dc motor drive by control of its armature current. Also, the term E_1/ω_s is proved to be mutual flux linkages as follows:

$$\frac{V_{as}}{\omega_s} \cong \frac{E_1}{\omega_s} = \frac{I_m X_m}{\omega_s} = \frac{I_m \omega_s L_m}{\omega_s} = I_m L_m = \lambda_m \tag{6.69}$$

where λ_m is the mutual flux linkages, much like the field-flux linkages produced by the current in the field coils of a dc machine. Note that this is fairly constant in the induction motor over a wide range of stator current. Equation (6.65) has given a powerful insight into the torque control of this motor drive, and it will be used in the closed-loop control described in the later section. That the induction motor control does not involve the control of all the three-phase currents, as in the case of a phase-controlled induction motor, is a significant point to be noted. Sensing only one current, e.g., dc link current, is sufficient for feedback control; this simplifies the control compared to any other ac drive scheme, as will become evident when other schemes are examined in later chapters.

Efficiency of the motor drive is calculated as follows :

$$\eta = \frac{\text{Shaft power output}}{\text{Power input}} = \frac{P_m - (\text{windage + friction losses})}{\text{Power input}} = \frac{P_m - P_{fw}}{P_i} \tag{6.70}$$

where P_{fw} is the windage and friction losses and P_i is the input power to the machine. It is calculated as the sum of the output power, rotor and stator resistive losses, core losses, and stray losses and is given by

$$P_i = P_m + 3[I_r^2(R_r + R_f) + I_c^2 R_c + I_s^2 R_s] + P_{st} \text{ (W)} \tag{6.71}$$

where

$$P_m = P_a(1 - s) = 3[I_{rl}R_{rf} - an_tV_{as}\cos\alpha]\cdot\frac{1-s}{s}\cdot I_{rl} \qquad (6.72)$$

The stator current is

$$I_{as} = I_{rl} + I_m + I_c = |I_{as}| \angle \phi_s \qquad (6.73)$$

Angle ϕ_s is the stator phase angle. The system input-phase current is the sum of the stator input current and the real and reactive components of the current supplied to the phase-controlled inverter from the lines through the step-down transformer. This is written as

$$\begin{aligned} I_{all} &= I_{as} - j(n_tI_{rrl}\sin\alpha) + n_tI_{rrl}\cos\alpha \\ &= I_{as}(\cos\phi_s + j\sin\phi_s) - j(an_tI_{rl}\sin\alpha) + an_tI_{rl}\cos\alpha = |I_{all}| \angle \phi l \end{aligned} \qquad (6.74)$$

where ϕ_l is the input-line power-factor angle of the utility mains supply.

In all these derivations, it must be noted that the harmonic current losses in the rotor have been neglected. This could be accounted for with the harmonic equivalent circuit. Equation (6.74) clearly shows that a lower transformer-turns ratio can improve the line power factor by keeping ϕ_l low.

The performance characteristics of the slip-energy-recovery-controlled induction motor drive are obtained by the use of the developed equivalent circuit and equations outlined above. For the purpose of solving the equivalent circuit, it could further be simplified by moving its magnetizing branch ahead of the stator impedance. This is shown in Figure 6.20.

A flowchart for the computation of performance characteristics is shown in Figure 6.21. A set of performance characteristics is shown in Figure 6.22. The motor and system details are as follows:

75 hp, 460 V, 3 ph, 60 Hz, 4 pole, Class D, $R_s = 0.0862\ \Omega$, $R_r = 0.9\ \Omega$, $L_{ls} = 0.78$ mH, $L_{lr} = 1.3$ mH, $L_m = 0.03$ H, $R_c = 150\ \Omega$, $R_f' = 0.04\ \Omega$, $a = 2.083$, $n_t = 0.6$.

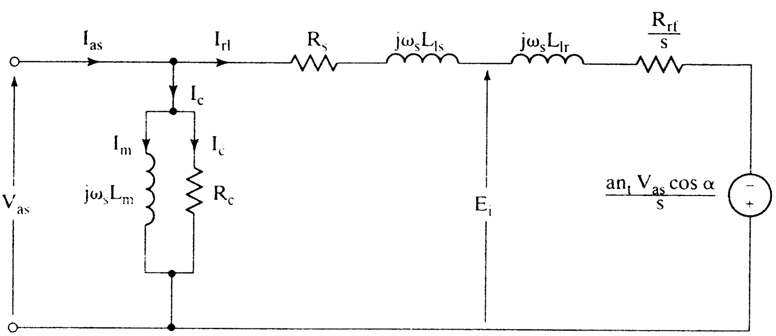


Figure 6.20 Simplified equivalent circuit of the slip-energy-recovery-controlled induction motor drive

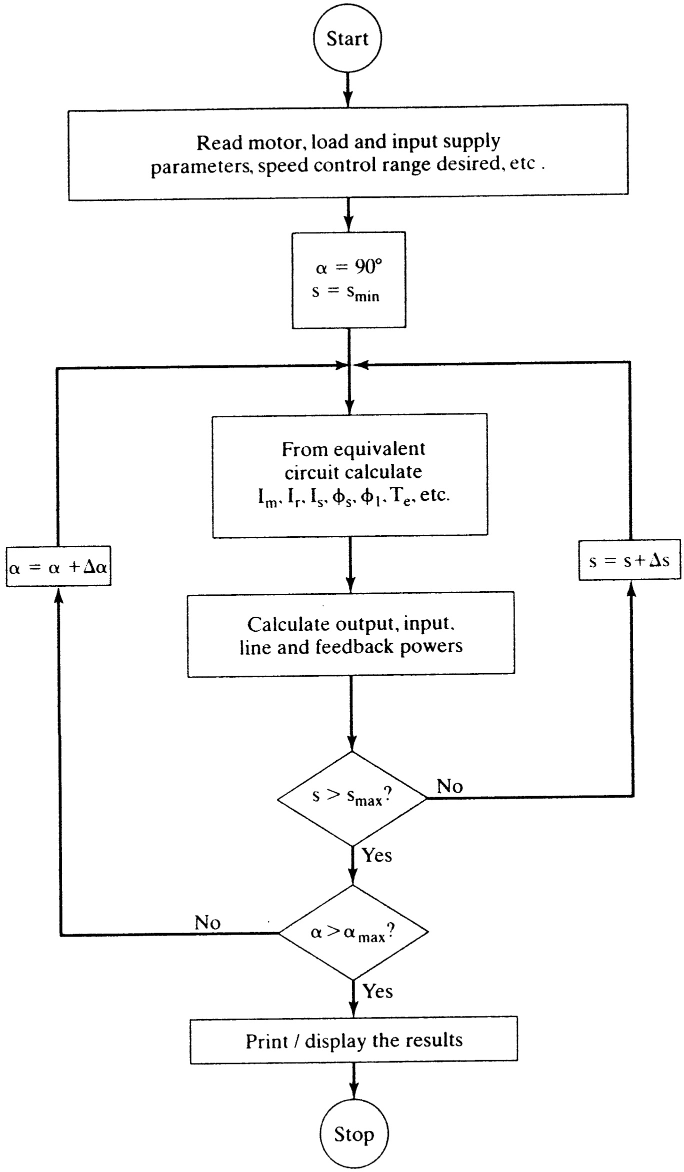
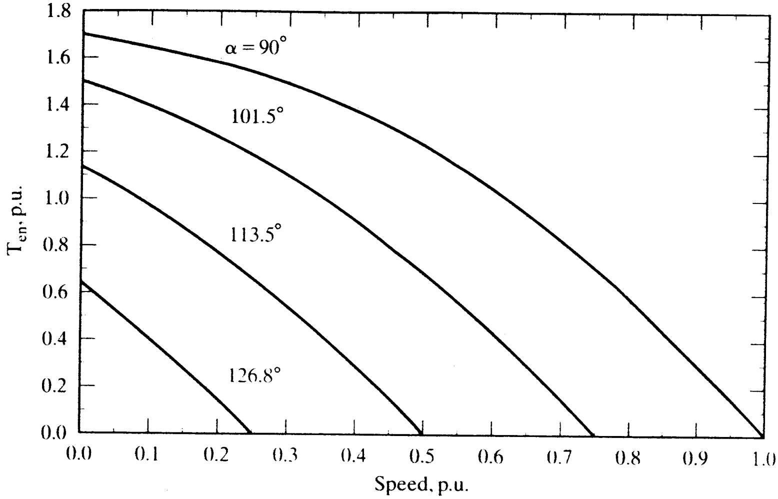
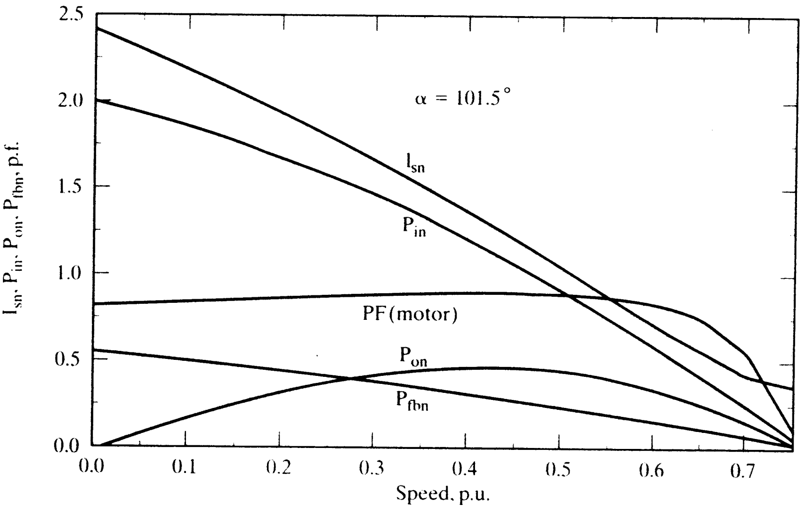


Figure 6.21 Flowchart for the computation of the slip-energy-recovery-controlled induction motor drive's steady-state performance



(i) Normalized torque vs. speed



(ii) System variables vs. speed

Figure 6.22 Performance characteristics of a slip-energy-recovery-controlled induction motor drive

The following procedure is used to compute I_{r1}. Write the voltage-loop equation involving the rotor emf source from Figure 6.19 (ii) as

$$\left\{\left(R_s + \frac{R_{rf}}{s}\right)I_{r1} - \frac{an_tV_{as}\cos\alpha}{s}\right\} + j\omega_s(L_{1s} + L_{1r})I_{r1} = V_{as} \qquad (6.75)$$

It must be noted that the variable emf source in the rotor is in phase with rotor current; hence, they are kept together as a unit in the expression. Considering only the magnitude part of the equation, the equation (6.75) is squared on both sides, and rearrangement yields a quadratic expression in I_{r1} as follows:

$$h_1I_{r1}^2 - h_2I_{r1} + h_3 = 0 \qquad (6.76)$$

where

$$h_1 = \left(R_s + \frac{R_{rf}}{s}\right)^2 + X_{eq}^2 \qquad (6.77)$$

$$X_{eq} = \omega_s(L_{1s} + L_{1r}) \qquad (6.78)$$

$$h_2 = \left\{\frac{2an_tV_{as}\cos\alpha}{s}\right\}\left\{R_s + \frac{R_{rf}}{s}\right\} \qquad (6.79)$$

$$h_3 = \left(\frac{an_tV_{as}\cos\alpha}{s}\right)^2 - V_{as}^2 \qquad (6.80)$$

The stator-referred fundamental rotor-current magnitude is obtained from equation (6.76) as

$$I_{r1} = \frac{h_2 + \sqrt{h_2^2 - 4h_1h_3}}{2h_1} \qquad (6.81)$$

The positive sign only is considered in the solution of I_{r1} in equation (6.81). Since h_2 is negative, only the addition of the discriminant yields a positive value for I_{r1}. Further, the following condition has to be imposed for the solution of I_{r1}:

$$\text{Real}\,\{I_{r1}\} > 0 \text{ or } I_{r1} = 0 \qquad (6.82)$$

For various triggering angles, the normalized electromagnetic torque-vs.-speed characteristics are drawn by using the procedure developed above and shown in Figure 6.22 (i). For a triggering angle of 101.5 degrees, the motor power factor, stator current of the motor, output power, P_{on}, input power, P_{in}, and feedback power from the rotor (which is the slip recovery power, P_{fbn}) are shown in Figure 6.22 (ii). Note that the motor power factor remains high for a wide speed range. As the speed decreases, the slip-energy recovery increases as well as the input power.

Example 6.3

Compare the efficiency of the slip-recovery-controlled induction motor drive with the maximum theoretical efficiency of the phase-controlled induction motor drive. The load torque is proportional to square of the speed. It is assumed that, at base speed, base torque is delivered. Consider operation at 0.6 p.u. speed. The details of the induction machine and its drive are given in the previous illustration of the torque-vs.-slip characteristics derivation.

Solution The following calculations from step 1 to step 5 are for the slip-recovery-controlled induction motor drive.

Step 1: Motor filter parameter for use in equivalent circuit is calculated.

$$R_{ff} = 0.5 * R_f' = 0.5 * 0.04 = 0.02\ \Omega$$
$$R_f = a^2 R_{ff} = 2.083^2 \times 0.02 = 0.0868\ \Omega$$
$$R_{rf} = R_r + R_f = 0.9 + 0.0868 = 0.9868\ \Omega$$

Step 2: Compute base values and load constant.

$$\text{Base speed, } \omega_b = \frac{2\pi f_s}{P/2} = \frac{2\pi 60}{4/2} = 188.5\ \text{rad/s}$$

$$\text{Base torque, } T_b = \frac{P_b}{\omega_b} = \frac{75 * 745.6}{188.5} = 296.66\ \text{N}\cdot\text{m}$$

$$\text{Load constant, } B_l = \frac{T_b}{\omega_b^2} = \frac{296.66}{188.5^2} = 0.0083\ \text{N}\cdot\text{m}/(\text{rad/s})^2$$

Step 3: Find the triggering angle and load torque.

Rotor speed = 0.6 p.u.

Slip = 1 − speed = 1 − 0.6 = 0.4

$$\alpha = \cos^{-1}\left(-\frac{s}{an_t}\right) = \cos^{-1}\left(-\frac{0.4}{2.083 * 0.6}\right) = 1.89\ \text{rad}$$

$$\text{Load torque, } T_l = B_l\omega_m^2 = 0.0083 * (0.6 * \omega_b)^2 = 106.8\ \text{N}\cdot\text{m}.$$

Step 4: Solve for rotor current from the air gap-torque expression. It must be noticed that electromagnetic torque is equal to the sum of the load torque and friction and windage torque. As the latter part is not specified in the problem, they are considered to be zero; hence, the electromagnetic torque is equal to the load torque in present case.

$$T_e = 3\frac{P}{2}\frac{1}{s\omega_s}\left[I_{r1}^2 R_{rf} - I_{r1} a n_t V_{as}\cos\alpha\right]$$

The only unknown in this expression is rotor current; it is solved for, yielding two values. Considering only the positive value, it is found as

$$I_{r1} = 21.12\ \text{A}$$

Step 5: From rotor current, the air gap emf is computed, from which all the other currents are evaluated. In order to do that, the reference phasor is assumed to be the rotor current. From the other currents, the stator and core losses are evaluated, and, from the rotor current, the rotor losses are calculated.

$$E_1 = \left[\frac{R_{rf}}{s} I_{r1} - \frac{a n_t V_{as}\cos\alpha}{s}\right] + j\omega_s L_{1r} I_{r1} = 158.3 + j10.35 = 158.67\angle 3.74^\circ$$

The various currents in the circuit are

Core loss current, $I_c = \dfrac{E_1}{R_c} = 1.05 + j0.069 = 1.06\angle 3.74°$

Magnetizing current, $I_m = \dfrac{E_1}{j\omega_s L_m} = 0.91 - j14.0 = 14.03\angle -86.26°$

No load current, $I_o = I_c + I_m = 1.97 - j13.93 = 14.07\angle -81.95°$

Stator phase current, $I_s = I_{r1} + I_o = 23.09 - j13.93 = 26.97\angle -31.1°$

Stator resistive losses, $P_{sc} = 3R_s I_s^2 = 3 * 0.0862 * 26.97^2 = 188.1$ W

Rotor and dc link filter resistive losses, $P_{rc} = 3R_{rf} I_{r1}^2 = 3 * 0.9868 * 21.12^2 = 1321$ W

Core losses, $P_{co} = 3R_c I_c^2 = 3 * 150 * 1.06^2 = 503.6$ W

Power output, $P_m = T_e \omega_m = 106.6 * (0.6 * 188.5) = 12{,}079$ W

Power input, $P_i = P_m + P_{sc} + P_{rc} + P_{co} = 14{,}091$ W

Efficiency, $\eta = \dfrac{P_m}{P_i} = \dfrac{12{,}079}{14{,}091} = 0.8572$

Step 6: Comparison of efficiency

Maximum theoretical efficiency possible with the phase-controlled induction motor drive, ignoring all stator, and core losses, is given as,

$$\eta = 1 - s = 1 - 0.4 = 0.6$$

In comparison to the slip-recovery-controlled induction-motor-drive efficiency, the efficiency of the phase-controlled drive system is 25.7% smaller. When the stator resistive and core losses are considered for the phase-controlled induction motor drive, the difference in efficiency will be even higher. It is left as an exercise to the reader.

6.3.4 Starting

The ratings of the transformer, rectifier bridge, and phase-controlled converter have to be equal to the motor rating if the slip-energy-recovery drive is started from standstill by using the controlled converter. The ratings of these subsystems can be reduced in a slip-energy-recovery drive with a limited speed-control region. In such a case, resistances are introduced in the rotor for starting. As the speed comes to the minimum controllable speed, these resistances are cut out and the rotor handles only a fractional power of the motor, thus considerably reducing the cost of the system. To withstand the heat generated in the resistors during starting, liquid cooling is recommended for large machines.

6.3.5 Rating of the Converters

Auxiliary starting means are assumed in the slip-energy-recovery control for the calculation of the converter rating. If an auxiliary means is not employed, the rating of the converters is equal to the induction-motor rating. As most of the applications for slip-energy-recovery drive are for fans and pumps, they have limited operational ranges of speed and hence power. The converters handle the slip power, and therefore their ratings are proportional to the maximum slip, s_{max}. The ratings of the converters are derived separately.

6.3.5.1 Bridge-rectifier ratings. The rectifier handles the slip power passing through the rotor windings. It is estimated in terms of air gap power as

$$\text{Slip power} = sP_a = s_{max}P_a = \text{Rectifier power rating, } P_{br} \tag{6.83}$$

where P_a is the air gap power and is given approximately, by neglecting the stator impedance and rotor leakage reactance, as

$$P_a \cong 3V_{as} I_{r1} \tag{6.84}$$

Hence, the ratings of the bridge rectifier are obtained by substituting (6.84) into (6.83):

$$P_{br} = s_{max}P_a = s_{max}\{3V_{as} \cdot I_{r1}\} \tag{6.85}$$

The peak voltage and rms current ratings of the diodes are obtained from the peak rotor line voltage and rms current in the rotor phase as

$$V_d = \sqrt{2}V_s = \sqrt{2}\sqrt{3}V_{as} = 2.45V_{as} \tag{6.86}$$

$$I_d = 0.816I_{dc} = 0.816 \times \frac{I_{rr1}}{0.779} = 1.05I_{rr1} = 1.05aI_{r1} \tag{6.87}$$

6.3.5.2 Phase-controlled converter. The peak voltage and rms current ratings of the power switches in the converter are

$$V_{ts} = \sqrt{2}\sqrt{3}n_tV_{as} = 2.45n_tV_{as} \tag{6.88}$$

$$I_{ts} = 0.816I_{dc} = 1.05aI_{r1} \tag{6.89}$$

6.3.5.3 Filter choke. The purpose of the dc-link filter choke is to limit the ripple in the dc link current. The ripple current is caused by the ripples in the rectified rotor voltages and phase-controlled inverter voltage reflected into the dc bus. Considering only the sixth harmonic gives the equivalent circuit of the dc link shown in Figure 6.23. The sixth-harmonic component of the rectified rotor voltages is written as

$$V_{dc6} = \sqrt{a_{dc6}^2 + b_{dc6}^2} \tag{6.90}$$

where

$$a_{dc6} = -\frac{6sV_s}{a\pi\sqrt{2}}\left[\frac{1}{7}\cos 7\alpha - \frac{1}{5}\cos 5\alpha\right] \tag{6.91}$$

$$b_{dc6} = \frac{6sV_s}{a\pi\sqrt{2}}\left[\frac{1}{7}\sin 7\alpha - \frac{1}{5}\sin 5\alpha\right] \tag{6.92}$$

By noting that the triggering angle for the diode bridge is zero, the sixth-harmonic voltage is evaluated as

$$V_{dc6} = \frac{0.077sV_s}{a} \tag{6.93}$$

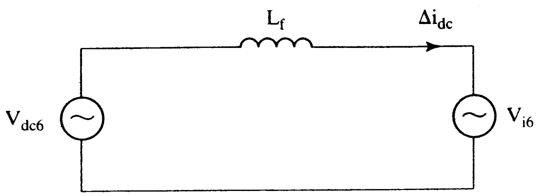


Figure 6.23 Sixth-harmonic equivalent circuit of the dc link

Similarly, the sixth-harmonic component of the inverter voltage is

$$V_{i6} = \frac{6n_tV_s}{\pi\sqrt{2}}\sqrt{\frac{1}{49} + \frac{1}{25} - \frac{2}{35}\cdot\cos 2\alpha} \tag{6.94}$$

and its maximum occurs at

$$\alpha = 90^\circ \tag{6.95}$$

giving the maximum of V_{i6} as

$$V_{i6m} = 0.463n_tV_s \tag{6.96}$$

These two harmonics are of different frequencies (i.e., v_{dc6} at six times slip frequency and V_{i6} at six times the line frequency) and have a phase shift between them, so it is safe to consider the sum of these voltages that has to be supported by the choke for the worst-case design. If the ripple current is denoted by ΔI_{dc},

$$L_f\frac{\Delta I_{dc}}{\Delta t} = V_{dc6} + V_{i6m} \tag{6.97}$$

where

$$\Delta t = \frac{2\pi}{6\omega_{sl}} \tag{6.98}$$

and ω_{sl} is the angular slip frequency. This is found to be lower than the supply frequency. The slip frequency preferably has to correspond to the maximum speed of the drive and hence the minimum slip speed.

By substituting for V_{dc6} and V_{i6m} from previous equations, the filter inductor is given as

$$L_f = \frac{1}{\Delta I_{dc}}\left[\frac{\pi}{6}\cdot\frac{V_s}{\omega_{sl}}\left\{0.463n_t + \frac{0.077s}{a}\right\}\right] \tag{6.99}$$

6.3.6 Closed-Loop Control

A closed-loop control scheme for torque and speed regulation is shown in Figure 6.24. The inner current loop is also the torque loop: the torque in a slip-energy-controlled drive is directly proportional to the dc link current. This torque loop is very similar to the phase-controlled dc motor drive's torque loop. The outer speed loop enforces the speed command, ω_r^*. In case of discrepancy between the speed and its

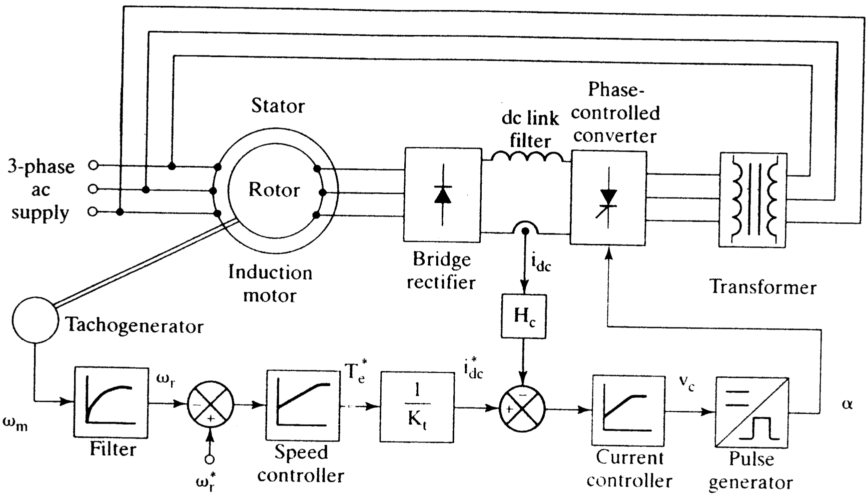


Figure 6.24 Closed-loop control of the slip-energy-recovery-controlled induction motor drive

command, the speed error is amplified and processed through a proportional-plus-integral controller. Its output forms the torque command. from which the current command is obtained by using the torque constant. The current error is usually processed by a proportional-plus-integral controller to produce a control voltage, v_c. This control voltage is converted into gate pulses with a delay of α degrees. The control block of pulse generators is explained in detail in Chapter 3. The only difference between the phase-controlled dc drive and this motor drive is that the triggering angle for motoring mode is usually less than 90° in the dc drive and greater than 90° in the slip-recovery drive. Accordingly, the control signal needs to be modified from the strategy used in the dc drive. This modification of the control circuit is very minor. The control signal v_c is limited so as to limit the triggering angle α to its upper limit of α_{max}, say, 155 degrees. The similarities between the dc drive and the slip-energy-controlled drive are exploited in the design of the current and speed controllers. The fact that there is only one control variable makes simple the design of controllers in the closed-loop system. As this is of one-quadrant operation, the PI controller outputs are zero for negative speed and current errors.

6.3.7 Sixth-Harmonic Pulsating Torques

The rotor currents in the induction motor are rich in harmonics. The effect of the dominant harmonics in the form of pulsating torque is examined in this section.

Writing the rectangular rotor current referred to the stator as a sum of Fourier components yields

$$I_r = \frac{2\sqrt{3}}{a\pi} I_{dc}\left[\cos\omega_s t - \frac{1}{5}\cos 5\omega_s t + \frac{1}{7}\cos 72\omega t - \frac{1}{11}\cos 11\omega_s t + \dots\right] \quad (6.100)$$

The harmonic rms currents are expressed in terms of the fundamental rms current from equation (6.100) as

$$I_m = \frac{1}{n} I_{r1}, \quad n = 5, 7, 11, 13, 17, 19, \ldots \tag{6.101}$$

where n is the harmonic number, and the fundamental stator-referred rotor current in terms of the dc link current is

$$I_{r1} = \frac{2\sqrt{3}}{a\pi\sqrt{2}} I_{dc} = \frac{0.779}{a} I_{dc} \tag{6.102}$$

Pulsating torques are produced when harmonic currents interact with the fundamental air gap flux and also when the harmonic air gap fluxes interact with the fundamental rotor current. The fundamental of the rotor current and air gap flux interact to produce a steady-state dc torque. The pulsating torques produce losses, heat, vibration, and noise, all of which are undesirable in the motor drive.

The predominant harmonics in the current are the fifth and the seventh, whose magnitudes are

$$I_{r5} = \frac{1}{5} I_{r1} \tag{6.103}$$

$$I_{r7} = \frac{1}{7} I_{r1} \tag{6.104}$$

Note that the fifth harmonic rotates at 5 times synchronous speed in the backward direction with reference to the fundamental. The seventh harmonic rotates at 7 times synchronous speed in the forward direction, as seen from equation (6.100). Hence, the relative speeds of the fifth and seventh harmonics with respect to the fundamental air gap flux are six times the synchronous speed. Similarly, the fifth- and seventh-harmonic air gap fluxes have a relative speed of six times the synchronous speed with respect to the fundamental of the rotor current. The harmonic air gap flux linkages are produced by the corresponding rotor harmonic currents. For example, the fifth-harmonic air gap flux linkages are

$$\lambda_{m5} = L_m I_{m5} = L_m \cdot \frac{E_5}{5X_m} = \frac{I_{r5}}{5\omega_s}\left\{\frac{R_r}{s_5} + j5X_{lr}\right\} \tag{6.105}$$

where the harmonic slip is given by

$$s_n = \frac{n\omega_s - \omega_r}{n\omega_s} = \frac{n\omega_s - \omega_s(1 - s)}{n\omega_s} = \frac{n - (1 - s)}{n} \tag{6.106}$$

For small fundamental slip *s*, the harmonic slip is given as

$$s_n \cong \frac{n \pm 1}{n}, \quad \begin{array}{l} + \text{ for } n = 5, 11, \ldots \\ - \text{ for } n = 7, 13, \ldots \end{array} \tag{6.107}$$

Hence, the fifth- and seventh-harmonic slips are obtained from the above as

$$s_5 = \frac{6}{5} = 1.2 \tag{6.108}$$

$$s_7 = \frac{6}{7} \tag{6.109}$$

Substituting for s_5 in equation (6.105), the fifth-harmonic air gap flux linkages are

$$\lambda_{m5} = \frac{I_{r1}}{25\omega_s}\left\{\frac{R_r}{1.2} + j5X_{lr}\right\} = \frac{I_{r1}}{30\omega_s}\{R_r + j6X_{lr}\} \tag{6.110}$$

Similarly, the seventh-harmonic air gap flux linkages are derived as

$$\lambda_{m7} = \frac{I_{r1}}{42\omega_s}\{R_r + j6X_{lr}\} \tag{6.111}$$

Because

$$6X_{lr} >> R_r \tag{6.112}$$

the fifth- and seventh-harmonic air gap flux linkages can be approximated as

$$\lambda_{m5} \cong \frac{I_{r1}}{5}L_{lr} \tag{6.113}$$

$$\lambda_{m7} \cong \frac{I_{r1}}{7}L_{lr} \tag{6.114}$$

Note that these flux linkages are leading the fundamental rotor current by 90 degrees. Representing the harmonic and fundamental variables and neglecting the leakage inductance effects, in Figure 6.25, leads to a calculation of the sixth-harmonic pulsating torques.

The fundamental of the electromagnetic torque is

$$T_{e1} = \frac{P_a}{\omega_s}\cdot\frac{P}{2} = \frac{3E_1I_{r1}\cos(\angle I_{r1} - \angle E_1)}{\omega_s}\cdot\frac{P}{2} = \frac{3\cdot\frac{P}{2}L_mI_{m1}\omega_sI_{r1}\sin\phi_{mr}}{\omega_s} = 3\cdot\frac{P}{2}\lambda_{m1}I_{r1}\sin\phi_{mr} \tag{6.115}$$

Because

$$\phi_{mr} = 90^\circ \tag{6.116}$$

when the rotor leakage inductance is neglected, the fundamental torque becomes

$$T_{e1} = 3\cdot\frac{P}{2}\lambda_{m1}I_{r1} \tag{6.117}$$

The sixth-harmonic pulsating torque is

$$\begin{aligned} T_{e6} &= 3\cdot\frac{P}{2}\{\lambda_{m1}(I_{r7} - I_{r5})\cos 6\omega_st + \lambda_{m5}I_{r1}\sin(90^\circ - 6\omega_st) + \lambda_{m7}I_{r1}\sin(90^\circ + 6\omega_st)\} \\ &= 3\cdot\frac{P}{2}\{\lambda_{m1}(I_{r7} - I_{r5}) + I_{r1}(\lambda_{m5} + \lambda_{m7})\}\cos 6\omega_st \end{aligned} \tag{6.118}$$

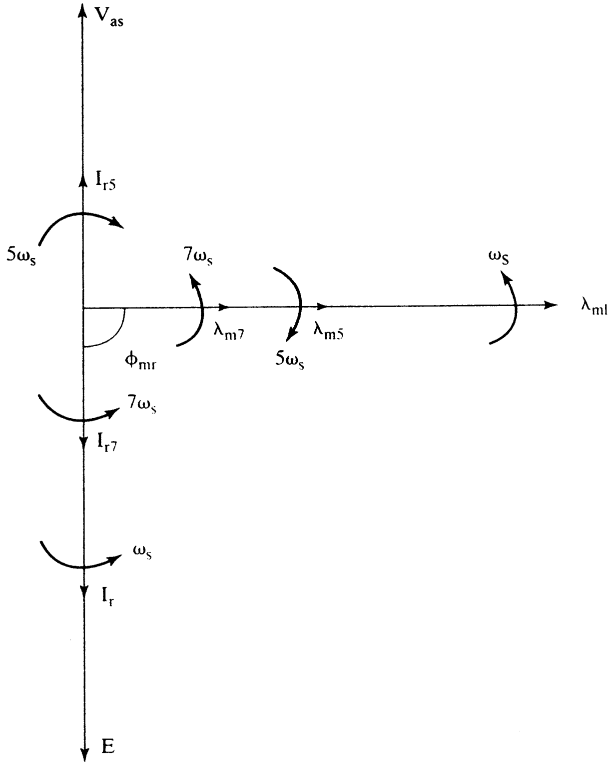


Figure 6.25 Phasor diagram of the rotor currents and air gap flux linkages

The maximum magnitude of the sixth-harmonic torque in terms of the fundamental torque is

$$\frac{|T_{e6}|}{T_{e1}} = \frac{(I_{r7} - I_{r5})}{I_{r1}} + \frac{I_{r1}L_{1r}\left(\frac{1}{5} + \frac{1}{7}\right)}{\lambda_{m1}} = -\frac{2}{35} + \frac{12}{35} \cdot \frac{I_{r1}L_{1r}}{\lambda_{m1}} \tag{6.119}$$

and the fundamental air gap flux linkage is approximated as

$$\lambda_{m1} \cong \frac{V_s}{\omega_s} \tag{6.120}$$

By substituting equation (6.120) into (6.119),

$$\frac{|T_{e6}|}{T_{e1}} = -\frac{2}{35} + \frac{12}{35} \cdot \frac{I_{r1}X_{1r}}{V_s} \tag{6.121}$$

For rated operating condition, the following assumptions are valid:

$$I_{r1} \cong 1 \text{ p.u.} \tag{6.122}$$

$$V_s = 1 \text{ p.u.} \tag{6.123}$$

and, hence,

$$\frac{|T_{e6}|}{T_{e1}} = -\frac{2}{35} + \frac{12}{35} \cdot X_{lr} \tag{6.124}$$

where X_{lr} is the p.u. leakage reactance. X_{lr} is normally less than 2% for large machines and less than 5% for small machines. From these values, it could be realized that the sixth-harmonic pulsating torque is very small and is inconsequential in fan and pump applications.

6.3.8 Harmonic Torques

As the fifth and seventh harmonics are of significant strength, it is necessary to consider and evaluate their own torques. Consider the magnitude of fifth-harmonic torque:

$$T_{e5} = 3 \cdot \frac{P}{2} \cdot \frac{I_{r5}^2 R_r (1 - s_5)}{s_5 \omega_s (1 - s)} \tag{6.125}$$

where

$$I_{r5} = \frac{I_{r1}}{5} \tag{6.126}$$

$$s_5 = \frac{-5\omega_s - \omega_s(1 - s)}{-5\omega_s} = \frac{6 - s}{5} \tag{6.127}$$

Note that the rotor time harmonics are referred to the stator for ease of computation. Substituting the equations (6.126) and (6.127) into equation (6.125) yields

$$T_{e5} = 3 \cdot \frac{P}{2} \cdot \frac{I_{r1}^2 R_r}{25\omega_s(s - 6)} \tag{6.128}$$

Expressing T_{e5} in terms of T_{e1} gives

$$\frac{T_{e5}}{T_{e1}} = \frac{s}{25(s - 6)} \tag{6.129}$$

The maximum magnitude of this occurs at a slip of one. Its value is

$$\max\left\{\frac{T_{e5}}{T_{e1}}\right\} = -0.008 \tag{6.130}$$

Note that the seventh-harmonic torque is much less and is of opposite sign. Hence, these harmonic torques can be ignored: they are of negligible magnitude.

6.3.9 Static Scherbius Drive

The slip-energy-recovery scheme described so far allows power to flow out of the induction-motor rotor and hence restricts the speed control to subsynchronous mode, i.e., below the synchronous speed. Regeneration is also not possible with this scheme. If the converter system is bidirectional, then both regeneration and super-synchronous modes of operation are feasible. This is explained as follows.

The air gap power is assumed to be a constant and is divided into the slip and mechanical power as given by the equation

$$P_a = P_{sl} + P_m \tag{6.131}$$

where

$$P_{sl} = sP_a \tag{6.132}$$

$$P_m = (1 - s)P_a \tag{6.133}$$

For a positive slip, the mechanical power is less than the air gap power, and the motor speed is subsynchronous. During this operation, the slip power is positive, and it is extracted from the rotor for feeding back to the supply. The drive system is in the subsynchronous motoring mode.

With the same assumption of constant air gap power, during regeneration the air gap power flows to the ac source. The torque is the input; hence, the mechanical power is considered to be negative. To maintain constant air gap power, the slip power becomes negative as seen from equation for negative air gap power. This results in input slip power, even though the slip remains positive. This mode of operation is subsynchronous regeneration.

For driving the rotor above the synchronous speed, the phase sequence of the rotor currents is reversed from that of the stator supply. This creates a field in the rotor opposite in direction to the stator field. The synchronous speed must be equal to the sum of the slip speed and rotor speed. Because the slip speed has become negative with respect to synchronous speed with the reversal of rotor phase sequence, the rotor speed has to increase beyond the synchronous speed by the amount of slip speed. The motor, thus, is in supersynchronous mode of operation, delivering mechanical power. In this mode, the mechanical power is greater than the air gap power. Note also that because of this the slip power is the input, as is seen from the power-balance equation of (6.131), and the slip is negative for supersynchronous operation.

To regenerate during supersynchronous operation, the slip power is extracted from the rotor. This becomes necessary, because the mechanical power is greater than the air gap power and hence the remainder constituting the slip power is available for recovery. Note that the slip remains negative and the rotor phase sequence is opposite to that of the stator. The present operation constitutes the supersynchronous regeneration mode. The various modes of operation are summarized in Table 6.2. A drive capable of all these modes is known as a Scherbius drive.

The converter system has to be bidirectional for the static Scherbius drive. Such a converter can be a cycloconverter, which is implemented with three sets of antiparallel three-phase controlled-bridge converters, discussed in Chapter 3. A

TABLE 6.2 Operational modes of static Scherbius drive

	Mode	Slip	Slip power	P_m
Subsynchronous	Motoring	+	+ (output)	$< P_a$
	Regeneration	+	− (input)	$< P_a$
Supersynchronous	Motoring	−	−	$> P_a$
	Regeneration	−	+	$> P_a$

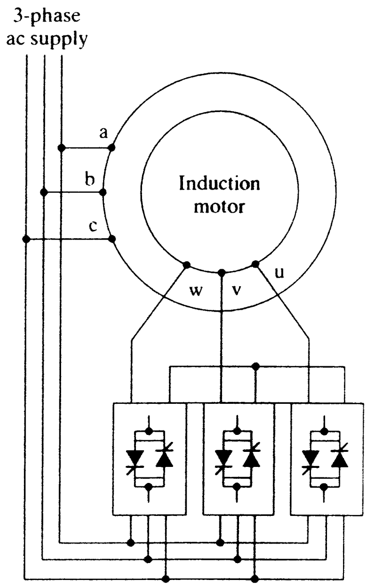


Figure 6.26 Static Scherbius drive

simple implementation of it in a block-diagram schematic is shown in Figure 6.26. This converter has a number of advantages:

(i) simple line commutation;
(ii) shaping of current, and hence power-factor improvement;
(iii) high efficiency;
(iv) suitable and compact for MW applications;
(v) capable of delivering a dc current (and hence the induction motor can be operated as a synchronous motor).

6.3.10 Applications

Slip-energy-recovery motor drives are used in large pumps and compressors in MW power ratings. It is usual to resort to an auxiliary starting means to minimize the

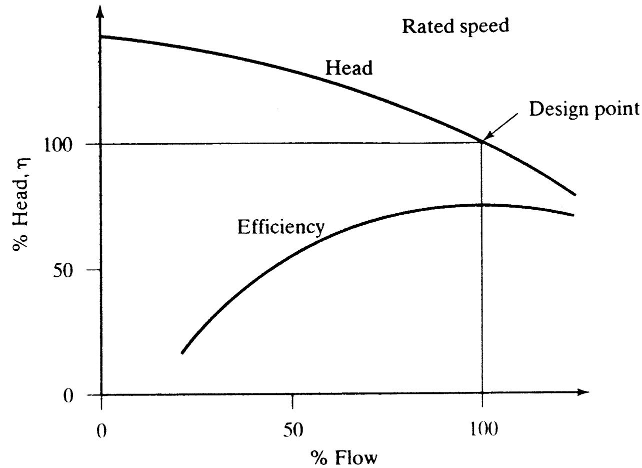


Figure 6.27 Head and efficiency vs. flow of pumps

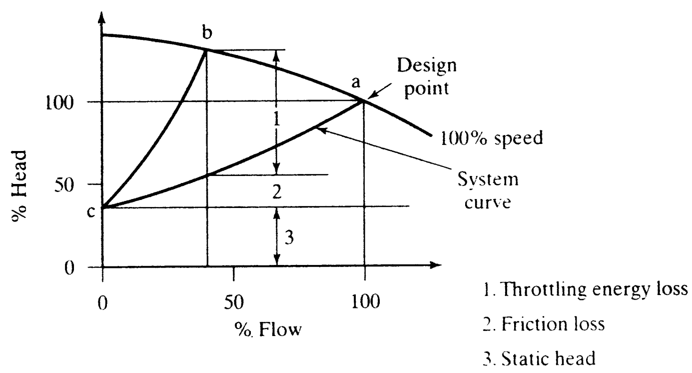


Figure 6.28 Flow reduction by damper control

converter rating. Wind-energy alternative power systems also use a slip-recovery scheme, but the converter rating cannot be minimized, because of the large variation in operational speed. An illustrative example of such an application can be found in reference [15]. The application is briefly described here.

The pump characteristics at rated speed are shown in Figure 6.27. The nominal operating point is at 1 p.u. flow, where the efficiency is maximum. If the flow has to be reduced, then the characteristic of the pump is changed by adding a resistance to the flow in the form of damper control, shown in Figure 6.28. The new operating point, *b*, is obtained by changing the system curve to move from *ca* to *cb* by adding system resistance from a damper control. At this operating point, the input energy to the pump will not change in proportion to the flow change, resulting in poor effi-

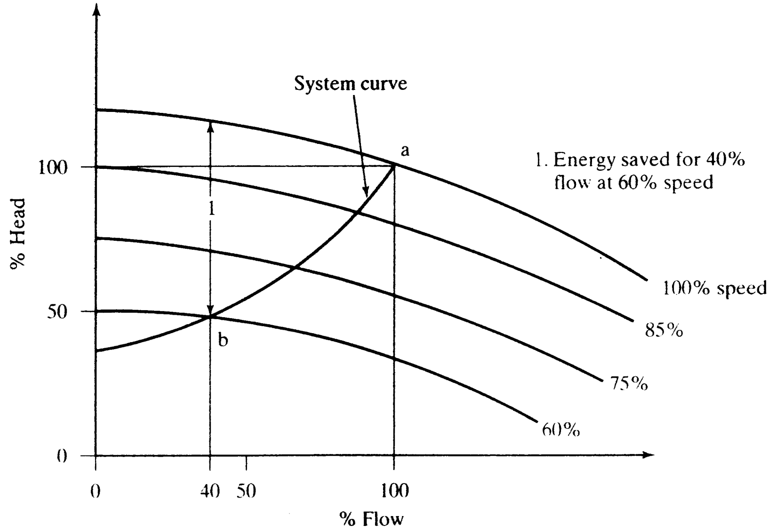


Figure 6.29 Flow control by speed variation

ciency because the input has to supply the throttling energy loss and friction energy loss. The flow can also be reduced if the pump characteristics are changed by reducing the pump speed with a variable-speed-controlled induction motor drive, and then the operating point moves as shown in Figure 6.29. The operating point can be always attempted on the highest-efficiency point by varying the speed. As the speed is reduced, the power input to the pump reduces and hence the input power to the driving motor decreases, resulting in higher energy savings compared to the damper control. Additionally, the combined efficiency of the motor and pump also improves by this method, resulting in considerable energy savings. Because the normal variation in the flow is between 60 and 100% of the rated value, the range of the variable speed control required usually is between 60 and 100% of the rated speed. Restricted-speed control requirement makes the slip-energy-recovery-controlled induction motor drive an ideal choice for this kind of application in the MW power range. It is not unusual to encounter, for example, 9.5-MW drive systems with a speed range of from 1135 to 1745 rpm and 11.5-MW drive systems with a speed range of from 757 to 1165 rpm, many of these in parallel or in series in many pumping stations. Such high-power drives get supplied from a 13.8 kV bus and are equipped with harmonic suppression circuits in the front end of the inverter and large dc reactors for smoothing in the dc link. They will have starting circuits with liquid-cooled resistors to reduce the power capability of the converters, and the inverter will be dual three-phase SCR-bridge converters in series, to keep the displacement factor above 0.88. Note tnat the displacement factor, which is the power factor considering only fundamentals, will be poor with a single three-phase bridge converter in the system. The converter units are force cooled by air or water in many installations.

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6.5 DISCUSSION QUESTIONS

1. Is there a difference in gating/triggering control between the induction-motor phase controller and the phase-controlled rectifier used in the dc motor drives?
2. In the phase-controlled induction motor drive, there could be 3 or 2, 2 or 1, 1 or 0 phases conducting, depending on the load. Envision the triggering angles and conduction patterns in these modes of operation.
3. The power switches can be connected in a number of ways to the star- and delta-connected stator of the induction motors. Enumerate the variations.
4. Is there an optimal configuration of the phase controller to be used in a delta-connected stator?
5. A single-phase capacitor-start induction motor is to have a phase controller. Discuss the best arrangement of its connection to the motor winding.
6. Discuss the configuration of the phase controller for a capacitor-start and a capacitor-run single-phase induction motor.
7. Discuss the merits and demerits of using a phase controller as a step-down transformer.
8. The relationship between the triggering angle and the fundamental of the output rms voltage is nonlinear in a phase-controlled induction motor drive. How can it be made linear? Is there any advantage to making it linear?
9. The phase controller is placed in series with the rotor windings as shown in Figure 6.30. Discuss the following:
   (i) the advantages of this scheme over the stator-phase-controlled induction motor drive;
   (ii) the current and voltage ratings of the power switches and their comparison to the stator-phase controller;

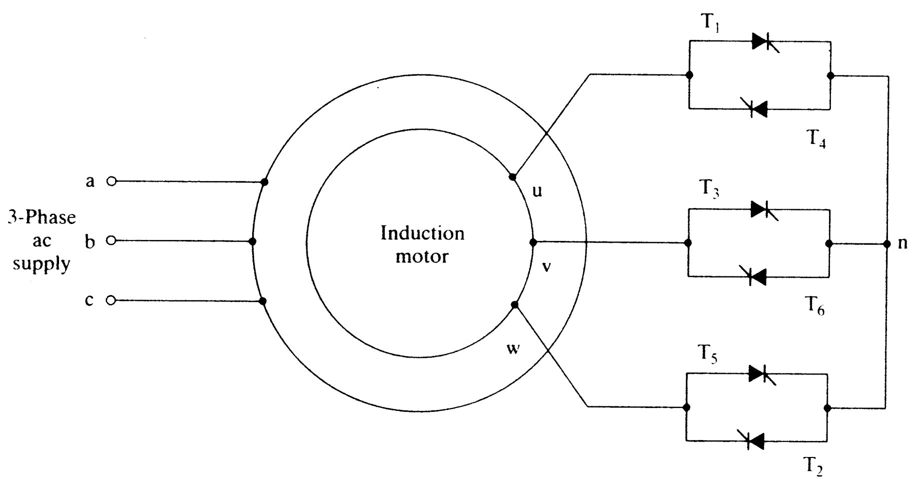

Figure 6.30 Rotor-phase-controlled induction motor drive

(iii) the improvement in the input power factor;
(iv) the improvement in line-current waveform and the reduction of harmonics;
(v) the disadvantages of this scheme;
(vi) the reversibility of the motor drive;
(vii) the control-signal generation for this scheme;
(viii) any soft-start possibility;
(ix) closed-loop control and its requirements.

10. Similarly to a phase-controlled converter (used in dc motor drives), whose α is limited in the inversion mode, will there be a need to limit the maximum value of α in the phase-controlled induction motor drive? Explain.
11. Compare the performance of the rotor-phase- and chopper-controlled induction motor drives.
12. Discuss the performance, analysis, design, merits, and demerits of the rotor chopper-controlled induction motor drive shown in Figure 6.31. (Hint: References 15 and 16.)
13. Compare the stator-phase and slip-energy-recovery-controlled induction motor drives on the basis of
    (i) input power factor
    (ii) efficiency
    (iii) cost
    (iv) control complexity
    (v) pulsating torque (6^{th} harmonic)
    (vi) feedback control
    (vii) range of speed control
14. What is the effect of the inductor in the dc link?
15. Suppose that the inductor in the dc link is replaced by an electrolytic capacitor. How would this affect the performance of the phase-controlled converter and consequently the slip-energy recovery?
16. The slip-energy-controlled drive, with a current loop controlling the dc link current, behaves like a separately-excited dc motor. Will a change in the load affect the flux and hence the torque constant?

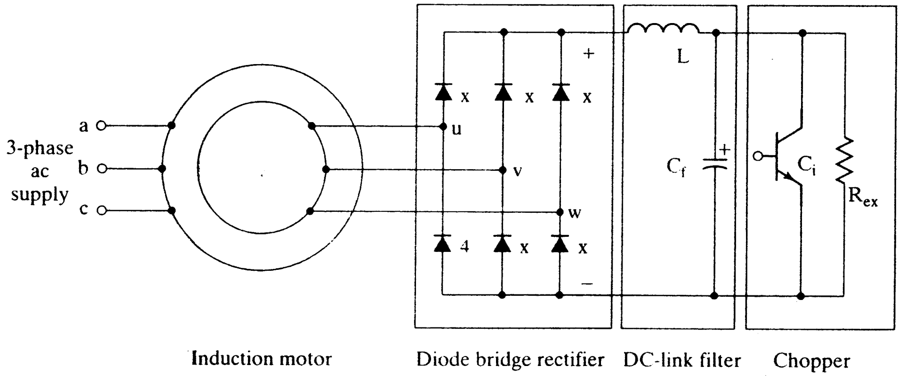

Figure 6.31 Rotor chopper-controlled induction motor drive

17. Discuss the feasibility of replacing the thyristor bridge converter with a GTO- or transistor-based converter to control the harmonics fed to the utility.
18. The diode-bridge rectifier in the dc link can be replaced by a GTO converter. By suitable control of this controlled converter, is it possible to control the current to eliminate the 6^{th} harmonic in the dc link and reduce the size of the dc inductor?
19. At the time of starting (assuming that there is no auxiliary starting means), what is the value of the triggering angle in the phase-controlled converter?
20. The following applications need a variable-speed motor drive:
    (i) a 20,000-hp pump;
    (ii) a 50-hp fan.

    The options available are stator-phase- and slip-energy-recovery-controlled induction motor drives. Which one is to be recommended for the above applications? Explain the reasons for your choice.
21. For open-loop speed control, NEMA class D induction motors are preferred in phase-controlled motor drives. Is the same choice valid for open-loop slip-energy-recovery-controlled-induction motor drives?
22. What are the modifications required in the controller of the phase-controlled converter used in a dc motor drive for adaptation to the slip-energy-recovery-controlled induction motor drive?
23. Can a slip-energy-controlled induction motor be reversed in speed? How is it done?

6.6 EXERCISE PROBLEMS

1. Determine the range of speed control obtained with a phase-controlled induction motor drive. The details are given below.

   1 hp, 2 pole, 230 V, 3 phase, star connected, 60 Hz, $R_s = 1\ \Omega$, $R_r = 6.4\ \Omega$, $X_m = 75\ \Omega$, $X_{lr} = 3.4\ \Omega$, $X_{ls} = 3\ \Omega$, and full-load slip is 0.1495. Load is a fan requiring 0.5 p.u. torque at 3200 rpm. α is limited to a maximum of 135 degrees.
2. Calculate and draw the efficiency of the drive given in Example 1 as a function of rotor speed.
3. A 1000-hp induction motor is to be started with a phase controller. The current is limited to 1 p.u. Find the starting torque and the range of triggering-angle variation to run the motor from standstill to 0.9 p.u. speed at a constant load torque of 0.05 p.u. The motor details are as follows:

   Base kVA = 850 kVA, Base voltage = 2700 V, Rated power = 1000 hp, Rated frequency = 50 Hz, P = 4, $R_s = 0.041\ \Omega$, $R_r = 0.1663\ \Omega$, $X_m = 80\ \Omega$, $X_{lr} = 0.2\ \Omega$, $X_{ls} = 0.2\ \Omega$, full-load slip = 0.0801 (base speed = full load speed).
4. Calculate the impact of rotor resistance increase by 80% on the speed-control range and efficiency of the drive given in problem 3.
5. The problem given in Example 1 is driving a fan load. At 1500 rpm, the load torque is 100%. Determine the open-loop stable operating-speed range for the triggering angles of $45° < \alpha < 135°$.
6. Determine approximately the magnitude of the two predominant harmonics present in the phase-controlled induction motor drive. Are these harmonics load-dependent?
7. A movie theater has been reconverted to a house one-fourth of its original seating. The ventilation fans have to be accordingly derated to reduce the operational costs. Phase

CHAPTER 7

Frequency-Controlled Induction Motor Drives

7.1 INTRODUCTION

The speed of the induction motor is very near to its synchronous speed, and changing the synchronous speed results in speed variation. Changing the supply frequency varies the synchronous speed. The relationship between the synchronous speed and the frequency is given by

$$n_s = \frac{120f_s}{P} \tag{7.1}$$

where n_s is the synchronous speed in rpm, f_s is the supply frequency in Hz, and P is the number of poles. The utility supply is of constant frequency, so the speed control of the induction motor requires a frequency changer to change the speed of the rotor. The frequency changers, circuit configurations, operation and input sources are discussed briefly in this chapter. The interaction of the frequency changers and the induction motor is studied in detail. They pertain to the steady-state performance, harmonics and their impact on the performance, control of harmonics, and drive-control strategies and their impact on drive performance. The enhancement of motor-drive efficiency is an integral aspect of the study and is considered with illustrative examples. The dynamic models of the various drive systems are derived and their performance simulated and analyzed as they pertain to design aspects.

7.2 STATIC FREQUENCY CHANGERS

There are basically two types of static frequency changers: direct and indirect. The direct frequency changers are known as cycloconverters: they convert an ac supply of utility frequency to a variable frequency. The cycloconverter for a single phase

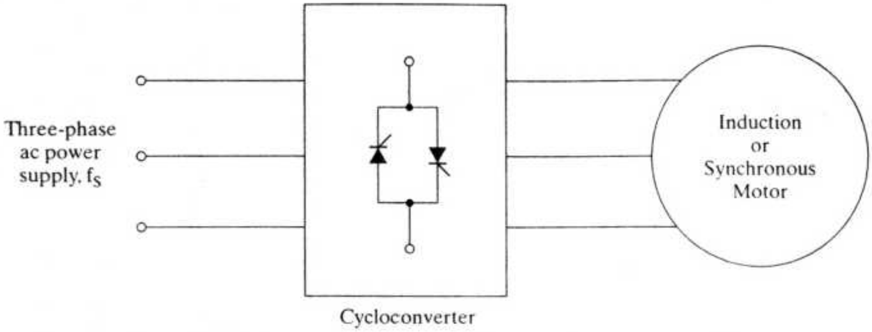

Figure 7.1 Cycloconverter-driven induction or synchronous motor drive

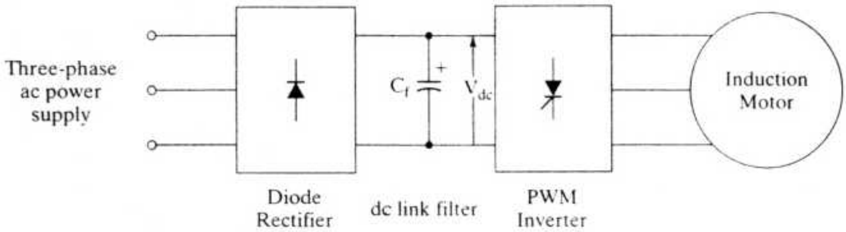

Figure 7.2 PWM inverter fed induction motor drive

consists of an antiparallel dual-phase-controlled converter. It can be a single- or three-phase-controlled converter; the latter version is very common in practice. Symbolically, a cycloconverter-driven ac motor drive is shown in Figure 7.1. The output frequency has a range of from 0 to $0.5f_s$. For better waveform control of the output voltage, the frequency is limited to $0.33f_s$. The smaller range of frequency variation is suitable for low-speed, large-power applications, such as ball mills and cement kilns. For a majority of applications, a wide range of frequency variation is desirable. In that case, indirect frequency conversion methods are appropriate. The indirect frequency changer consists of a rectification (ac to dc) and an inversion (dc to ac) power conversion stage. They are broadly classified depending on the source feeding them: voltage or current sources. In both these sources, the magnitude should be adjustable. The output frequency becomes independent of the input supply frequency, by means of the dc link.

If rectification is uncontrolled, the voltage and its frequency are controlled in the inverter; a pulse-width-modulated inverter-fed induction motor drive is shown in Figure 7.2. The dc link filter consists of a capacitor to keep the input voltage to the inverter constant and to smooth the ripples in the rectified output voltage. The dc link voltage cannot reverse; it is a constant, so this is a voltage-source drive. The advantage of using the diode-bridge rectifier in the front end is that the input line power factor is nearly unity, but there is also a disadvantage, in that the power cannot be recovered from the dc link for feeding back into the input supply. The regen-

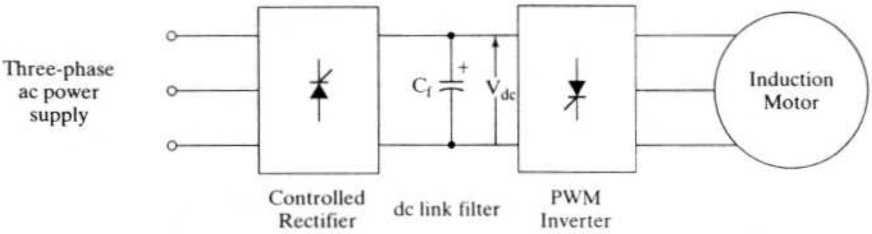

Figure 7.3 Variable-voltage, variable-frequency (VVVF) induction motor drive

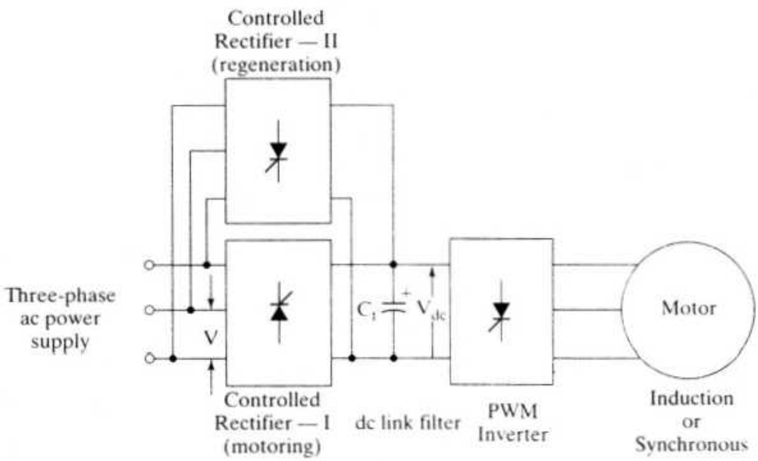

Figure 7.4 Regenerative voltage-source inverter-fed ac drive

erative power has to be handled by some other controller arrangement and is discussed later.

Separating the magnitude and frequency control functions in the controlled rectifier and inverter, respectively, gives a configuration shown in Figure 7.3. This configuration is known as a variable-voltage, variable-frequency induction motor drive. It has a disadvantage: the power factor is low at low voltages, from phase control. Refer to Chapter 3 on the operation of the phase-controlled rectifier. To recover the regenerative energy in the dc link, the direction of the dc link current to the phase-controlled rectifier has to be reversed; the dc link voltage cannot reverse through the antiparallel diodes across the inverter bridge. Hence, an antiparallel-controlled rectifier is required to handle the regenerative energy, as shown in Figure 7.4. Similarly to the voltage-source induction motor drives, the current-source drives have a PWM control and variable-current variable-frequency control.

In the PWM current source, shown in Figure 7.5, instantaneous current control on the ac side is enforced by a set of fast-acting current-control loops. Inverter switching is based on the current error signals to force the actual currents to track their

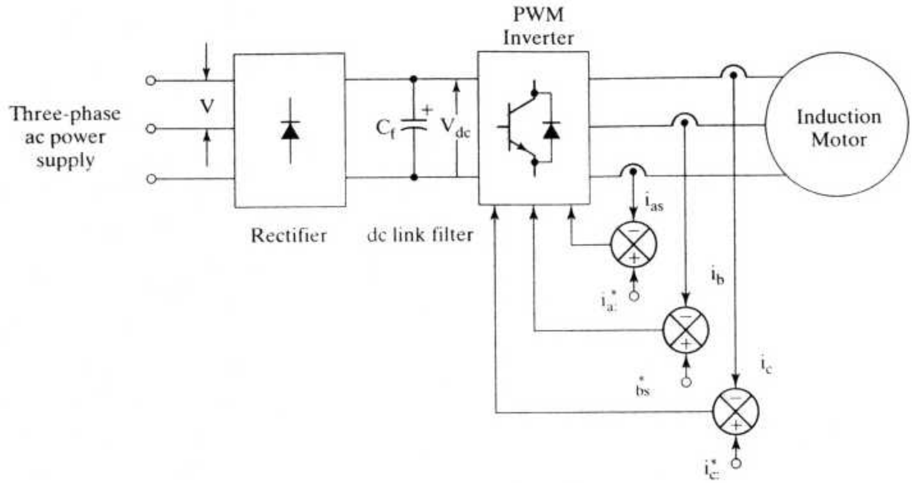

Figure 7.5 Current-regulated voltage-source-driven induction motor drive

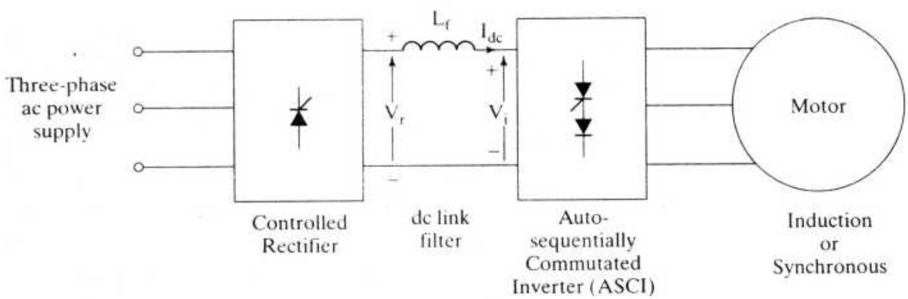

Figure 7.6 Current-source inverter-driven induction or synchronous motor drive

respective commanded values. The input source is voltage; because of this fact, this arrangement is sometimes referred to as a *current-regulated induction motor drive.*

In the variable-current, variable-frequency (VCVF) systems, the current magnitude and frequency control are exercised independently by the controlled rectifier with an inner current loop and autosequentially commutated inverter (ASCI), respectively. This arrangement is capable of four-quadrant operation and is discussed in detail in later subsections. To maintain a current source, the dc link filter has an inductor and a current loop enforcing the dc link current command, very much like the inner current loop in the phase-controlled dc motor drive. The current source drive is shown in Figure 7.6. The disadvantages of this drive scheme are the need for a large dc link inductor and a set of commutation capacitors. A general schematic of classification of the static frequency changers is shown in Figure 7.7.

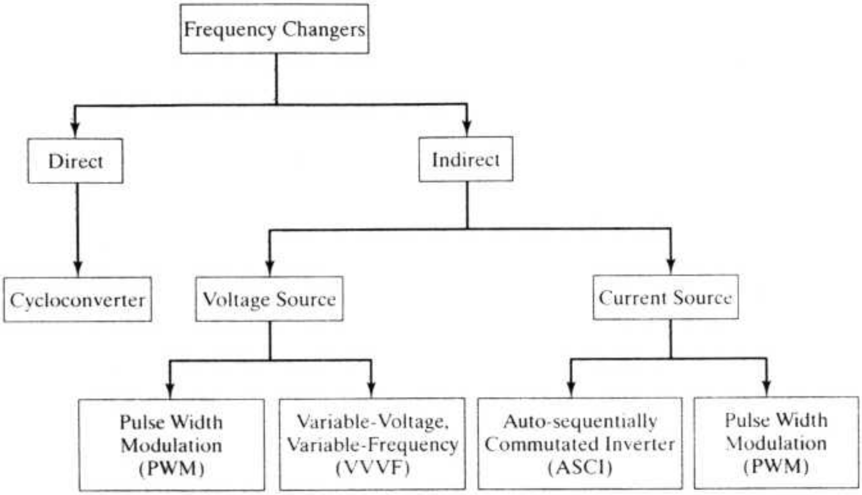

Figure 7.7 Classification of frequency changers

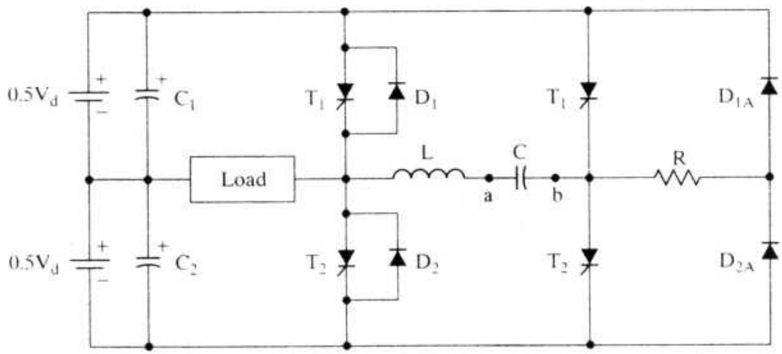

Figure 7.8 Half-bridge voltage-source inverter

7.3 VOLTAGE-SOURCE INVERTER

A commonly used SCR inverter is the modified McMurray inverter in industrial motor drives. The configuration and principle of operation of this inverter are explained in this section. For ease of development, a half-bridge inverter is considered. Then it is extended to include a full-bridge inverter. The transistorized inverter is discussed, and a comparison between SCR and transistor inverters is given that is based on the number of switches.

7.3.1 Modified McMurray Inverter

A half-bridge modified McMurray inverter is shown in Figure 7.8. The inductor L and capacitor C are known as the commutating inductor and capacitor, respectively.

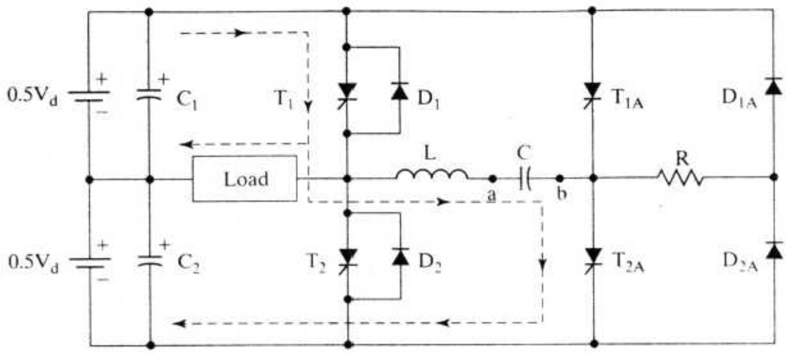

Figure 7.9a (a) Half-bridge voltage-source inverter charging

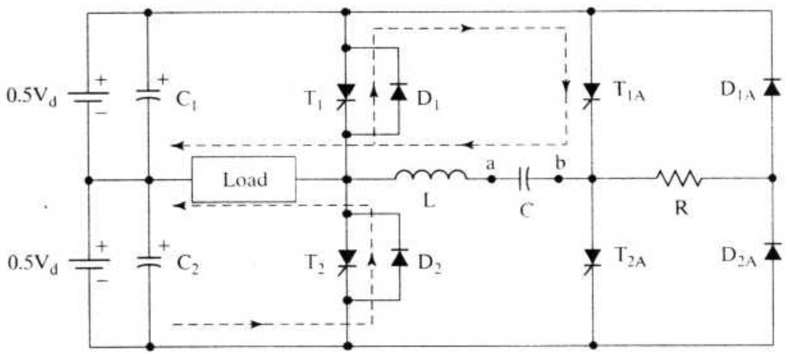

Figure 7.9b (b) Half-bridge voltage-source inverter commutation

This inverter accepts a dc voltage (source) and provides an alternating voltage of variable frequency across its load. Its operation is described as follows.

The circuit arrangement assumes that the snubbers are placed across the devices to limit dv/dt and its effects. At the time of starting, let the voltage across the commutating capacitor C be zero.

Step 1: T_1 and T_{2A} are gated on. The sequence of operation is shown in Figure 7.9(a). A current is established from the dc source which partly goes through switch T_1 and the load; the rest goes through T_1, L, C, and T2A. The capacitor C is charged with *a* positive and *b* negative. The capacitor will charge to a voltage greater than twice the source voltage (V_d) in an LC circuit. The polarity of *b* and the anode of T2Abecome very much negative to the cathode of T2A, thus turning off the device. Note that the current through the load is maintained. The excess charge in the capacitor due to its overvoltage will forward bias D_1 if load current is zero or flows through the load and D_{2A}, circulating a current through the bottom source voltage $0.5V_d$, L, C, and R. When the capacitor voltage is equal to the source voltage, this current ceases.

Step 2: To turn T_1 off, T_{1A} is turned on as shown in Figure 7.9(b), thus applying a positive voltage from capacitor C to the cathode of T_1, forcing it to cease conduc-

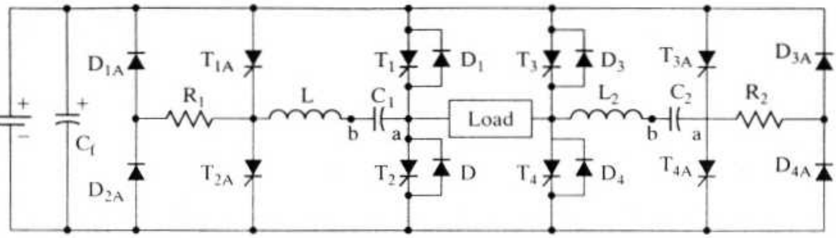

Figure 7.10 Full-bridge voltage-source inverter

tion. An oscillation through L, C, T_1, and T_{1A} occurs. When the capacitor current exceeds load current, the current in T_1 completely ceases. The current in excess of load current flows through D_1 now. This maintains a negative voltage across T_1, to let it recover to withstand forward voltage. The voltage across C is reversed during commutation. Assuming that the load is reactive, the load current will be in the same direction, thus forcing D_2 to be forward biased. Because of the action of diode D_2, the load voltage will become negative. Similarly, the negative-half-cycle output-voltage operation can be realized.

The frequency of the load voltage is determined by the rate at which T_1 and T_2 are enabled. This half-bridge inverter does not fully utilize the source voltage. A half-bridge inverter is placed on each side of the load to make a full-bridge inverter, shown in Figure 7.10.

7.3.2 Full-Bridge-Inverter Operation

The main thyristors are commutated by the auxiliary thyristors, which are denoted by an additional A in their subscripts. The operation of this circuit is explained here.

Step 1: T_1, T_{2A}, T_4, and T_{3A} are gated on. C_1 and C_2 are charged with *a* positive and *b* negative. Load voltage and load current are positive. The excess charge in C_1 is drained through V_d, D_{2A}, R_1, L_1, C_1, and D_1 if there is no load current and through the load if there is a load current. The excess charge in C_2 is drained through V_d, D_4, L_2, C_2, R_2, and D_{3A}.

Step 2: To provide a zero voltage across the load for part of the positive half cycle, turn T_{1A} on and thus turn T_1 off. The load is shorted via D_2, load, and T_4, thus driving the voltage to zero. The zeroing of load voltage is used to change the effective volt–sec across the load.

Step 3: T_1 and T_4 are turned off by turning on T_{1A} and T_{4A}, respectively, to prepare for the negative half-cycle across the load. The priming of commutating capacitors and gating on of T_2 and T_3 take place in a manner similar to the positive-half-cycle operation described in steps 1 and 2.

Note that this single-phase full-bridge inverter is identical to the four-quadrant chopper described in Chapter 4, except that the present one is an SCR version. A three-phase inverter is made of three such single-phase inverters.

A chopper with self-commutating power switches, such as a transistor, has been studied in Chapter 4. It is recalled that a four-quadrant chopper is also a single-phase

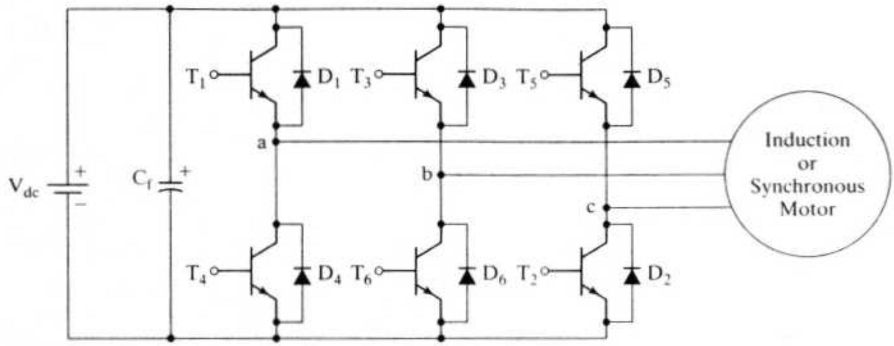

Figure 7.11 Voltage-source inverter with transistors

TABLE 7.1 Comparison of three-phase SCR and self-commutating inverters

Comparison Aspect	SCR Inverter	Self-Commutating Inverter
SCRs/Transistors	24	6
Diodes	24	6
Inductors/Capacitors	6/6	–
Fundamental RMS output phase voltage	$0.78V_d$	$0.45V_d$

inverter. A combination of three of the two-quadrant choppers, i.e., with two power switches, results in a three-phase inverter. The voltage available to one phase is always less than the full source or dc link voltage in such a case. A three-phase self-commutating inverter with transistor switches is shown in Figure 7.11. Note the simplicity of the configuration and the minimum use of power devices: the transistors are self-commutating. A comparison of power switches and other factors for the SCR and self-commutating inverters is given in Table 7.1. Note that the auxiliary thyristors are not rated to be equal to the main thyristors: they operate only for a fraction of time compared to the main SCRs. The same argument applies equally to the rating of the auxiliary diodes.

7.4 VOLTAGE-SOURCE INVERTER-DRIVEN INDUCTION MOTOR

7.4.1 Voltage Waveforms

A generic self-commutating three-phase inverter is shown in Figure 7.12 in schematic, with its output connected to the three stator phases of a wye-connected induction motor. The gating signals and the resulting line voltages are shown in Figure 7.13. The power devices are assumed to be ideal: when they are conducting, the voltage across them is zero; they present an open circuit in their blocking mode. The phase voltages are derived from the line voltages in the following manner by assuming a balanced three-phase system.

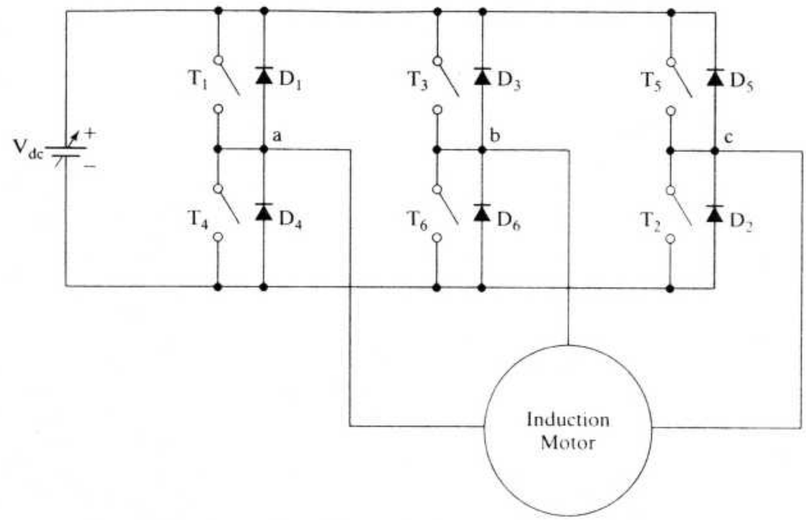

Figure 7.12 A schematic of the generic inverter-fed induction motor drive

The line voltages in terms of the phase voltages in a three-phase system with phase sequence *abc* are

$$V_{ab} = V_{as} - V_{bs} \tag{7.2}$$

$$V_{bc} = V_{bs} - V_{cs} \tag{7.3}$$

$$V_{ca} = V_{cs} - V_{as} \tag{7.4}$$

where V_{ab}, V_{bc}, and V_{ca} are the various line voltages and V_{as}, V_{bs}, and V_{cs} are the phase voltages. Subtracting equation (7.4) from equation (7.2) gives

$$V_{ab} - V_{ca} = 2V_{as} - (V_{bs} + V_{cs}) \tag{7.5}$$

In a balanced three-phase system, the sum of the three phase voltages is zero:

$$V_{as} + V_{bs} + V_{cs} = 0 \tag{7.6}$$

Using equation (7.6) in (7.5) shows that the difference between line voltages V_{ab} and V_{ca} is

$$V_{ab} - V_{ca} = 3V_{as} \tag{7.7}$$

from which the phase *a* voltage is given by

$$V_{as} = \frac{V_{ab} - V_{ca}}{3} \tag{7.8}$$

Similarly, the *b* and *c* phase voltages are

$$V_{bs} = \frac{V_{bc} - V_{ab}}{3} \tag{7.9}$$

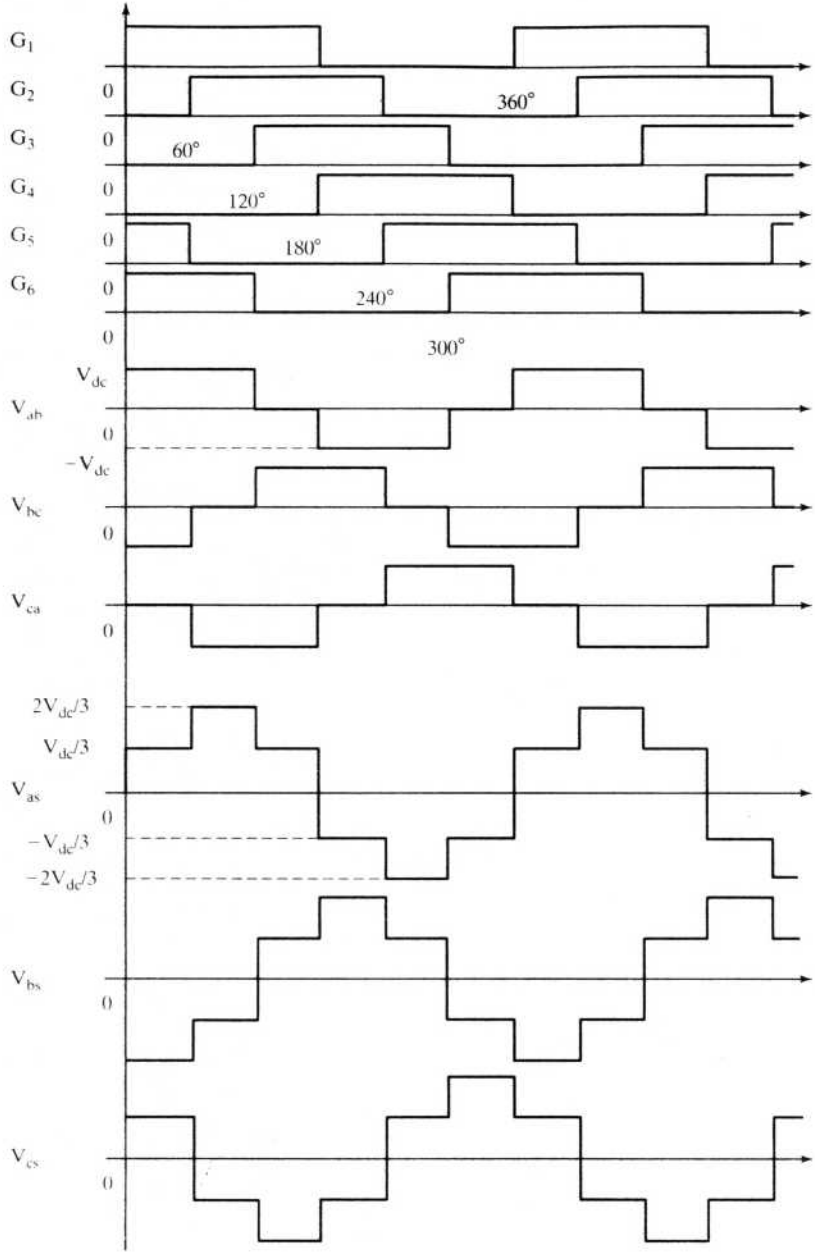

Figure 7.13 Inverter gate (base) signals and line- and phase-voltage waveforms

$$V_{cs} = \frac{V_{ca} - V_{bc}}{3} \tag{7.10}$$

The phase voltages derived from line voltages are shown in Figure 7.13. Although the line-to-line voltages are 120 electrical degrees in duration, the phase voltages are six-stepped and of quasi-sine waveforms. These periodic voltage waveforms, when resolved into Fourier components, have the following form:

$$v_{ab}(t) = \frac{2\sqrt{3}}{\pi}V_{dc}\left(\sin \omega_s t - \frac{1}{5}\sin 5\omega_s t + \frac{1}{7}\sin 7\omega_s t - \dots\right) \tag{7.11}$$

$$v_{bc}(t) = \frac{2\sqrt{3}}{\pi}V_{dc}\left\{\sin(\omega_s t - 120°) - \frac{1}{5}\sin(5\omega_s t - 120°) + \frac{1}{7}\sin(7\omega_s t - 120°) - \dots\right\} \tag{7.12}$$

$$v_{ca}(t) = \frac{2\sqrt{3}}{\pi}V_{dc}\left\{\sin(\omega_s t + 120°) - \frac{1}{5}\sin(5\omega_s t + 120°) + \frac{1}{7}\sin(7\omega_s t + 120°) - \dots\right\} \tag{7.13}$$

The phase voltages are shifted from the line voltages by 30 degrees, and their magnitudes are $\frac{2}{\pi}V_{dc}$. Only the fundamental produces useful torque, and hence only it needs to be considered for the steady-state performance evaluation of inverter-fed ac motor drives. In this regard, the fundamental rms phase voltage for the six-stepped waveform is

$$V_{ph} = \frac{V_{as}}{\sqrt{2}} = \frac{2}{\pi}\cdot\frac{V_{dc}}{\sqrt{2}} = 0.45V_{dc} \tag{7.14}$$

7.4.2 Real Power

The dc link transfers real power to the inverter and induction motor. Assuming that the harmonic powers are negligible, and considering only the fundamental input power to the induction motor,

$$P_i = V_{dc}I_{dc} = 3V_{ph}I_{ph}\cos\phi_1 \tag{7.15}$$

where I_{dc} is the average steady dc link current, I_{ph} is the phase current, and ϕ_1 is the fundamental power-factor angle in the induction motor. Substituting for V_{ph} from equation (7.14) into (7.15), we get

$$I_{dc} = 1.35I_{ph}\cos\phi_1 \tag{7.16}$$

7.4.3 Reactive Power

The induction motor requires reactive power for its operation. Since the dc link does not supply the reactive power, this reactive power is supplied by the inverter. It is done by the switching in the inverter, and hence the inverter can be considered a reactive power generator. The facility to turn on and off the phase currents allows the inverter to vary the phase angle between the current and voltages. This ability has been endowed by solid-state power switching and has enormous impact on the control of ac machines. This subject is treated in subsequent sections and chapters. The fundamental input reactive power demand of the induction motor is

$$Q_i = 3V_{ph}I_{ph}\sin\phi_1 \tag{7.17}$$

The inverter is therefore rated for $3V_{ph}I_{ph}$ and in volt-amp, which is the apparent power. If the load power factor is low, then the inverter rating goes up.

7.4.4 Speed Control

Speed control is achieved in the inverter-driven induction motor by means of variable frequency. Apart from frequency, the applied voltage needs to be varied, to keep the air gap flux constant and not let it saturate. This is explained as follows.

The air gap induced emf in an ac machine is given by

$$E_1 = 4.44k_{\omega 1}\phi_m f_s T_1 \tag{7.18}$$

where $k_{\omega 1}$ is the stator winding factor, ϕ_m is the peak air gap flux, f_s is the supply frequency, and T_1 is the number of turns per phase in the stator. Neglecting the stator impedance, $R_s + jX_{ls}$, the induced emf approximately equals the supply-phase voltage. Hence,

$$V_{ph} \cong E_1 \tag{7.19}$$

The flux is then written as

$$\phi_m \cong \frac{V_{ph}}{K_b f_s} \tag{7.20}$$

where

$$K_b = 4.44k_{\omega 1}T_1 \tag{7.21}$$

If K_b is constant, flux is approximately proportional to the ratio between the supply voltage and frequency. This is represented as

$$\phi_m \propto \frac{V_{ph}}{f_s} \propto K_{vf} \tag{7.22}$$

where K_{vf} is the ratio between V_{ph} and f_s.

From equation (7.22), it is seen that, to maintain the flux constant, K_{vf} has to be maintained constant. Therefore, whenever stator frequency is changed to obtain speed control, the stator input voltages have to be changed accordingly to maintain the air gap flux constant. This particular requirement compounds the control problem and sets it apart from the dc motor control, which requires only the voltage control. The implications of such a requirement during the transients made the dc drives preferable over the ac drives until the 1970s.

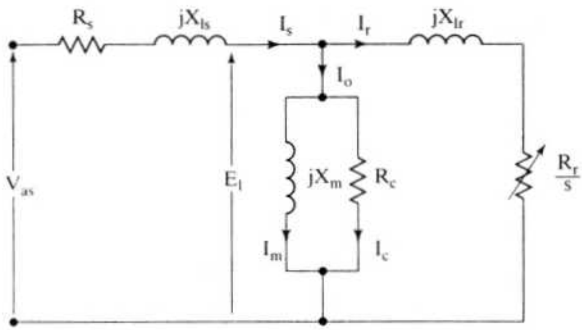

Figure 7.14 Equivalent circuit of the induction motor

A number of control strategies have been formulated, depending on how the voltage-to-frequency ratio is implemented:

(i) Constant volts / Hz control
(ii) Constant slip-speed control
(iii) Constant air gap flux control
(iv) Vector control

These control strategies are considered individually, the first three in the following sections; the fourth is reserved for the next chapter.

7.4.5 Constant Volts/Hz Control

7.4.5.1 Relationship between voltage and frequency. The applied phase voltage from the equivalent circuit shown in Figure 7.14 is

$$V_{as1} = E_1 + I_{s1}(R_s + jX_{ls}) \tag{7.23}$$

where I_{s1} is the fundamental stator phase current.

The dependence of the phase voltage on the stator impedance drop in p.u. is derived as follows.

$$\frac{V_{as1}}{V_b} = \frac{E_1}{V_b} + \frac{I_{s1}}{V_b}(R_s + jX_{ls}) \tag{7.24}$$

or

$$V_{asn} = E_{1n} + I_{s1n}(R_{sn} + jX_{lsn}) \tag{7.25}$$

where

$$V_{asn} = \frac{V_{as1}}{V_b}\ (\text{p.u.})$$

$$E_{1n} = \frac{E_1}{V_b} = \frac{j(L_m I_m)\omega_s}{\lambda_b \omega_b} = j\left(\frac{\lambda_m}{\lambda_b}\right)\left(\frac{\omega_s}{\omega_b}\right) = j\lambda_{mn}\,\omega_{sn} \tag{7.26}$$

$$I_{sn} = \frac{I_{sl}}{I_b}, \text{ p.u.} \tag{7.27}$$

$$R_{sn} = \frac{I_b R_s}{V_b}, \text{ p.u.} \tag{7.28}$$

$$X_{1n} = \frac{I_b L_{1s} \omega_s}{V_b} = L_{1sn}\omega_{sn}, \text{ p.u.} \tag{7.29}$$

$$\lambda_{mn} = \frac{\lambda_m}{\lambda_b}, \text{ p.u.} \tag{7.30}$$

The p.u. fundamental input phase voltage is written as

$$V_{asn} = I_{sn}R_{sn} + j\omega_{sn}(\lambda_{mn} + L_{1sn}I_{sn}) \text{ (p.u.)} \tag{7.31}$$

where L_{lsn} is the p.u. stator leakage inductance and ω_{sn} is the p.u. stator frequency. Substituting equation (7.31) into (7.30) gives the normalized input-phase stator voltage:

$$V_{asn} = \sqrt{(I_{sn}R_{sn})^2 + \omega_{sn}^2(\lambda_{mn} + L_{1sn}I_{sn})^2} \text{ (p.u.)} \tag{7.32}$$

where R_{sn} is the p.u. stator resistance and λ_{mn} is the p.u. airgap flux linkages.

For constant air gap flux linkages of 1 p.u., the p.u. applied voltage vs. p.u. stator frequency is shown in Figure 7.15. The stator resistance and leakage inductance are 0.026 and 0.051 p.u., respectively, for this graph. The graphs are shown for 0.25-, 0.5-, 1-, and 2-p.u. stator currents. Even though, for constant flux, the relationship between the applied voltage and the stator frequency appears linear, note that it

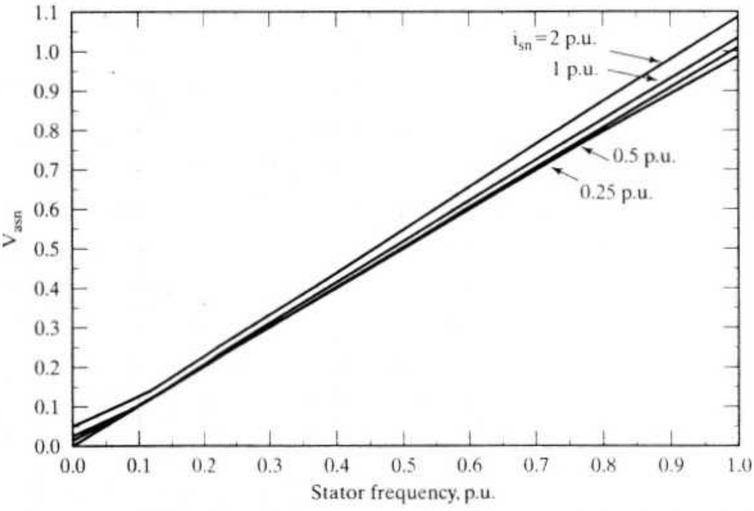

Figure 7.15 Normalized stator phase voltage vs. stator frequency for various stator currents

does not go through the origin; we need a small voltage to overcome the stator resistance at zero frequency.

Figure 7.15 demonstrates that the volt/Hz ratio needs to be adjusted in dependence on the frequency, the air gap flux magnitude, the stator impedance, and the magnitude of the stator current. Such a complex implementation is not desirable for low-performance applications, such as fans and pumps; there it is usual to have a preprogrammed volts-to-frequency relationship, as shown in Figure 7.16. This allows for an offset voltage at zero stator frequency to overcome the stator resistance drop, which is kept as an adjustable parameter in the inverters. The relationship between the applied phase voltage and frequency in general is written as

$$V_{as} = V_o + K_{vf} f_s \tag{7.33}$$

where

$$V_o = I_{sl} R_s \tag{7.34}$$

V_o is the offset voltage to overcome the stator resistive drop. This can be converted into the dc link voltage by converting equation (7.14) into normalized form and combining it with equation (7.33). It is carried out as follows.

$$V_{asn} = \frac{V_{ph}}{V_b} = 0.45 \times \frac{V_{dc}}{V_b} = 0.45 V_{dcn} \tag{7.35}$$

$$V_{on} = \frac{V_o}{V_b}\,(\text{p.u.})$$

$$E_{1n} = \frac{E_1}{V_b} = \frac{K_{vf} f_s}{K_{vf} f_b} = f_{sn}$$

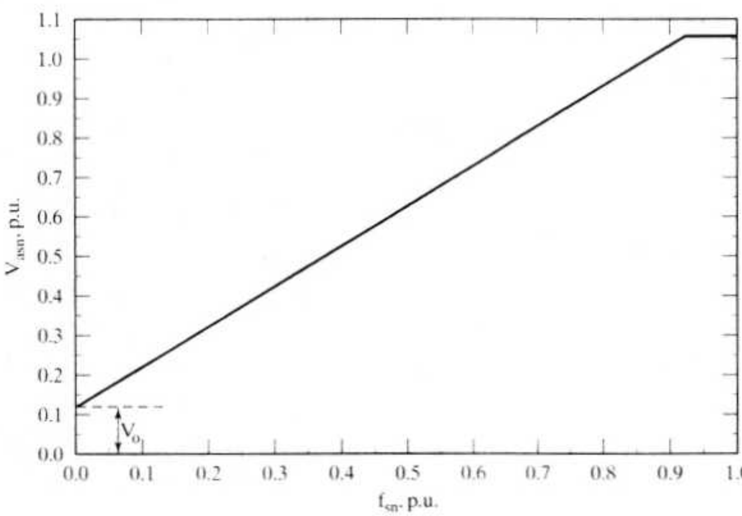

Figure 7.16 General implementation of the voltage-to-frequency profile in inverter-fed induction motor drives

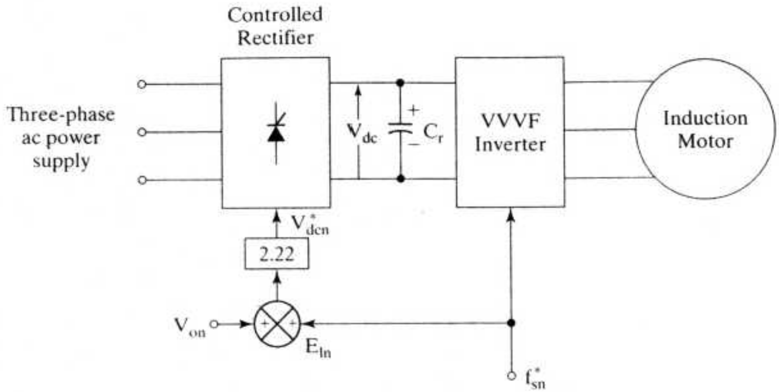

Figure 7.17 Implementation of volts/Hz strategy in inverter-fed induction motor drives

Hence,

$$0.45V_{dcn} = V_{on} + f_{sn} = V_{on} + E_{In} \tag{7.36}$$

or

$$V_{dcn} = 2.22\{V_{on} + f_{sn}\} \tag{7.37}$$

where V_{dcn} is the dc link voltage in p.u. and E_{In} is the normalized induced emf.

7.4.5.2 Implementation of volts/Hz strategy. An implementation of the constant volts/Hz control strategy for the inverter-fed induction motor in open loop is shown in Figure 7.17. This type of variable-speed drive is used in low-performance applications where precise speed control is not necessary. The frequency command f^*_{sn} is enforced in the inverter and the corresponding dc link voltage is controlled through the front-end converter. The offset voltage, V_{on}, is added to the voltage proportional to the frequency, and they are multiplied by 2.22 to obtain the dc link voltage.

Some problems encountered in the operation of this open-loop drive are the following:

1. The speed of the motor cannot be controlled precisely, because the rotor speed will be less than the synchronous speed. Note that stator frequency, and hence the synchronous speed, is the only variable controlled in this drive.
2. The slip speed, being the difference between the synchronous and electrical rotor speed, cannot be maintained: the rotor speed is not measured in this drive scheme. This can lead to operation in the unstable region of the torque–speed characteristics.
3. The effect discussed in 2 can make the stator currents exceed rated current by many times, thus endangering the inverter–converter combination.

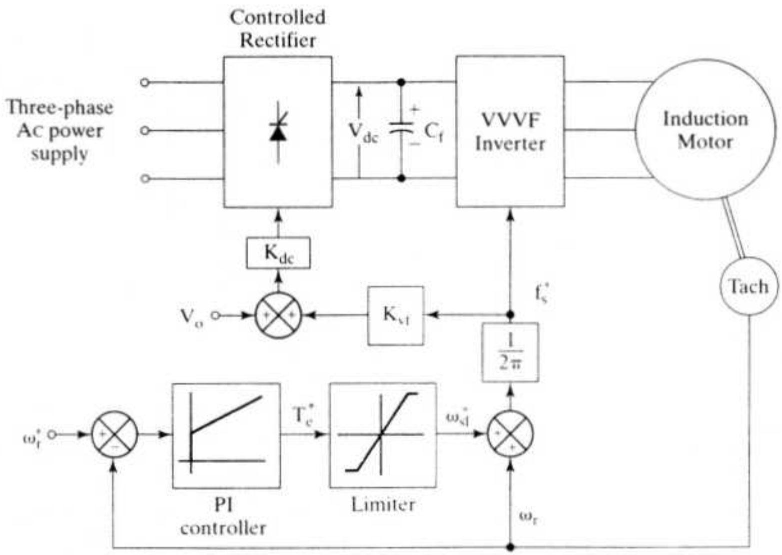

Figure 7.18 Closed-loop induction motor drive with constant volts/Hz control strategy

These problems are, to an extent, overcome by having an outer speed loop in the induction motor drive, shown in Figure 7.18. The actual rotor speed is compared with its commanded value, ω_r^*, and the error is processed through a controller, usually a PI, and a limiter to obtain the slip-speed command, ω_{sl}^*. The limiter ensures that the slip-speed command is within the maximum allowable slip speed of the induction motor. The slip-speed command is added to electrical rotor speed to obtain the stator frequency command. Thereafter, the stator frequency command is processed as in an open-loop drive. K_{dc} is the constant of proportionality between the dc load voltage and the stator frequency.

In the closed-loop induction motor drive, the limits on the slip speed, offset voltage, and reference speed are externally adjustable variables. This external adjustment allows the tuning and matching of the induction motor to the converter and inverter and the tailoring of its characteristics to match the load requirements.

Example 7.1

Find the relationship between the dc link voltage and the stator frequency for the closed-loop implementation of a volts/Hz inverter-fed induction motor drive. The motor parameters are as follows:

5 hp, 200 V, 60 Hz, 3 phase, star connected, 4 pole, 0.86 pf and 0.82 efficiency.

$$R_s = 0.277\ \Omega,\ R_r = 0.183\ \Omega,\ X_m = 20.30\ \Omega,\ X_{ls} = 0.554\ \Omega,\ X_{lr} = 0.841\ \Omega$$

Solution The constants required for the implementation of the closed-loop drive are: K_{vf}, V_o, and the maximum slip speed.

(i) Maximum slip speed $\cong \dfrac{R_r}{(L_{1r} + L_{1s})} \cong \dfrac{0.183}{(0.84 + 0.554)/377.0} = 49.5 \text{ rad / sec}$

(ii) $V_o = I_{sl}R_s$

where

$$I_{sl} = \text{Rated stator phase current} = \frac{\text{hp} \times 745.6}{3V_{ph} \times \text{pf} \times \text{efficiency}} = \frac{5 \times 745.6}{3 \times 115.5 \times 0.86 \times 0.82} = 15.26 \text{ A}$$

Hence

$$V_o = 15.26 \times 0.277 = 4.23 \text{ V}$$

(iii) $$K_{vf} = \frac{(V_{ph} - V_o)}{f_s} = \frac{\left(\dfrac{200}{\sqrt{3}} - 4.23\right)}{60} = 1.854 \text{ volts/ Hz}$$

(iv) The dc link voltage in terms of stator frequency is given by

$$V_{dc} = 2.22V_{as} = 2.22\{V_o + K_{vf}f_s\} = 2.22\{4.23 + 1.854f_s\} = (9.4 + 4.12f_s), \text{V}$$

7.4.5.3 Steady-state performance. The steady-state performance of the constant-volts/Hz-controlled induction motor drive is computed by using the fundamental applied phase voltage given in the expression (7.33). In the equivalent circuit, the following steps are taken to compute the steady-state performance:

Step 1: Start with minimum stator frequency and zero slip.
Step 2: Calculate magnetizing, core-loss, rotor, and stator phase currents.
Step 3: Calculate the electromagnetic torque, power output, copper, and core losses.
Step 4: Calculate input power factor and efficiency.
Step 5: Increment the slip; then go to step 2, unless $s = s_{max}$.
Step 6: Increment the stator frequency; then go to step 1, unless $f_s = f_{s\,max}$.

A set of drive-torque-vs.-speed characteristics is given in Figure 7.19 for the motor constants given in Example 7.1. The dissimilarity between the torque–speed characteristics for various stator frequencies is quite notable. The peak torque in the motoring region decreases as the stator frequency decreases, contrary to the generator action. Note that the offset voltage is set at zero for this figure. The effects of offset-voltage variations are shown in Figure 7.20 for the same motor at a stator frequency of 15 Hz. The offset voltage increases the induced emf and rotor current, resulting in enhanced torque.

7.4.5.4 Dynamic simulation. The dynamic simulation of the constant-volts/Hz-controlled induction motor drive is developed in this section. The models of the motor, inverter, dc link, controlled rectifier, and drive controllers are developed, and the equations are combined to obtain the transient response. As a

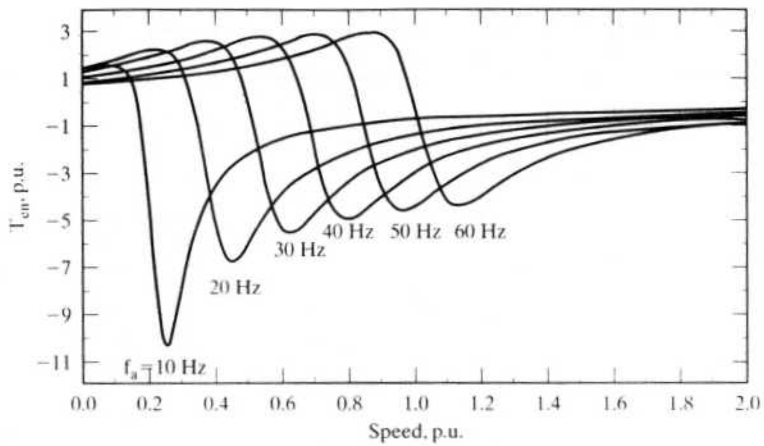

Figure 7.19 Torque–speed characteristics at various stator frequencies

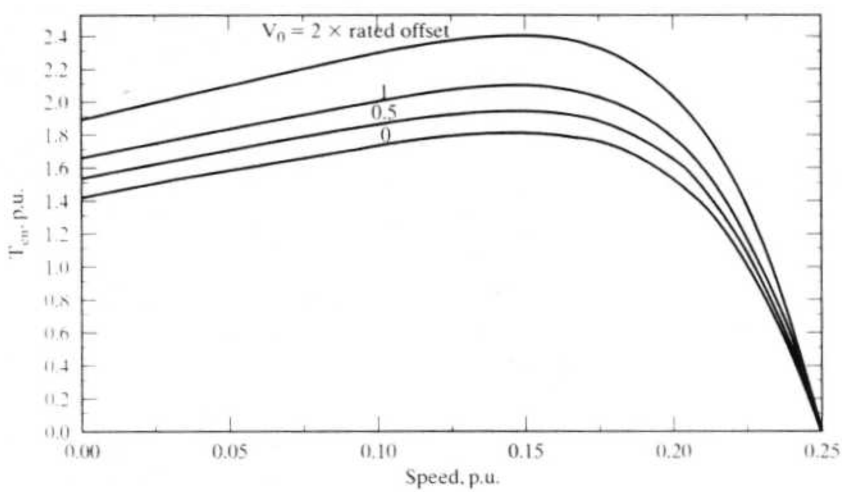

Figure 7.20 Torque-speed characteristics of the volts/Hz-controlled induction motor drive with various voltage offsets at f_s = 15 Hz

corollary, a direct method of evaluating steady-state performance without going through initial transients is also outlined. The simulation considers the actual voltage waveforms without ignoring harmonic contents in the applied voltages.

Motor: The induction motor model in synchronously rotating reference frames is chosen, because it gives many advantages. The same model after perturbations can be used for small-signal response, since the input and output variables become dc quantities (considering the fundamental alone).

The model is given in Chapter 5. The equations of the induction motor in synchronous reference frame are as follows:

$$\begin{bmatrix} v_{qs}^e \\ v_{ds}^e \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} R_s + L_s p & \omega_s L_s & L_m p & \omega_s L_m \\ -\omega_s L_s & R_s + L_s p & -\omega_s L_m & L_m p \\ L_m p & (\omega_s - \omega_r)L_m & R_r + L_r p & (\omega_s - \omega_r)L_r \\ -(\omega_s - \omega_r)L_m & L_m p & -(\omega_s - \omega_r)L_r & R_r + L_r p \end{bmatrix} \begin{bmatrix} i_{qs}^e \\ i_{ds}^e \\ i_{qr}^e \\ i_{dr}^e \end{bmatrix} \tag{7.38}$$

and the electromechanical system equation is

$$J\frac{d\omega_r}{dt} = \frac{P}{2}(T_e - T_l) - B\omega_r$$

and electromagnetic torque is

$$T_e = \frac{3}{2}\frac{P}{2}L_m(i_{qs}^e i_{dr}^e - i_{ds}^e i_{qr}^e) \tag{7.39}$$

The synchronous reference frames voltage vector is

$$v_{qdo}^e = [T_{abc}^e]v_{abc} \tag{7.40}$$

where the transformation from *abc* to *qdo* variables is given by

$$[T_{abc}^e] = \frac{2}{3}\begin{bmatrix} \cos\theta_s & \cos(\theta_s - 2\pi/3) & \cos(\theta_s + 2\pi/3) \\ \sin\theta_s & \sin(\theta_s - 2\pi/3) & \sin(\theta_s + 2\pi/3) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \tag{7.41}$$

$$v_{qdo}^e = [v_{qs}^e \quad v_{ds}^e \quad v_o]^t \tag{7.42}$$

$$v_{abc} = [v_{as} \quad v_{bs} \quad v_{cs}]^t \tag{7.43}$$

where the symbols are explained in Chapter 5.

Input voltages: The input phase voltages are the six stepped waveforms shown in Figure 7.13. From the transformation given in (7.40) to (7.43), the *d* and *q* stator voltages in synchronously rotating reference frames are as follows.

For the interval $0 \le \theta_s \le \frac{\pi}{3}$, the *d* and *q* axes voltages are

$$v_{qs1}^e(\theta_s) = \frac{2}{3}V_{dc}\cos\left(\theta_s + \frac{\pi}{3}\right) \tag{7.44}$$

$$v_{ds1}^e(\theta_s) = \frac{2}{3}V_{dc}\sin\left(\theta_s + \frac{\pi}{3}\right) \tag{7.45}$$

where

$$\theta_s = \omega_s t \tag{7.46}$$

and for the interval $\frac{\pi}{3} \le \theta_s \le \frac{2\pi}{3}$, the voltages are

$$v^e_{qsII}(\theta_s) = \frac{2}{3} V_{dc} \cos(\theta_s) \tag{7.47}$$

$$v^e_{dsII}(\theta_s) = \frac{2}{3} V_{dc} \sin(\theta_s) \tag{7.48}$$

but the equations (7.47) and (7.48) can be written as

$$v^e_{qsII}(\theta_s) = \frac{2}{3} V_{dc} \cos\left(\theta_s + \frac{\pi}{3} - \frac{\pi}{3}\right) \tag{7.49}$$

$$v^e_{dsII}(\theta_s) = \frac{2}{3} V_{dc} \sin\left(\theta_s + \frac{\pi}{3} - \frac{\pi}{3}\right) \tag{7.50}$$

Equations (7.44), (7.45), (7.49), and (7.50) signify that q and d axis voltages are similar for each 60° and that they are periodic. They are shown in Figure 7.21. The symmetry of these voltages is exploited to find the steady state directly. Note that this voltage representation considers the actual waveforms of the inverter; i.e., the fundamental and higher-order harmonics are included.

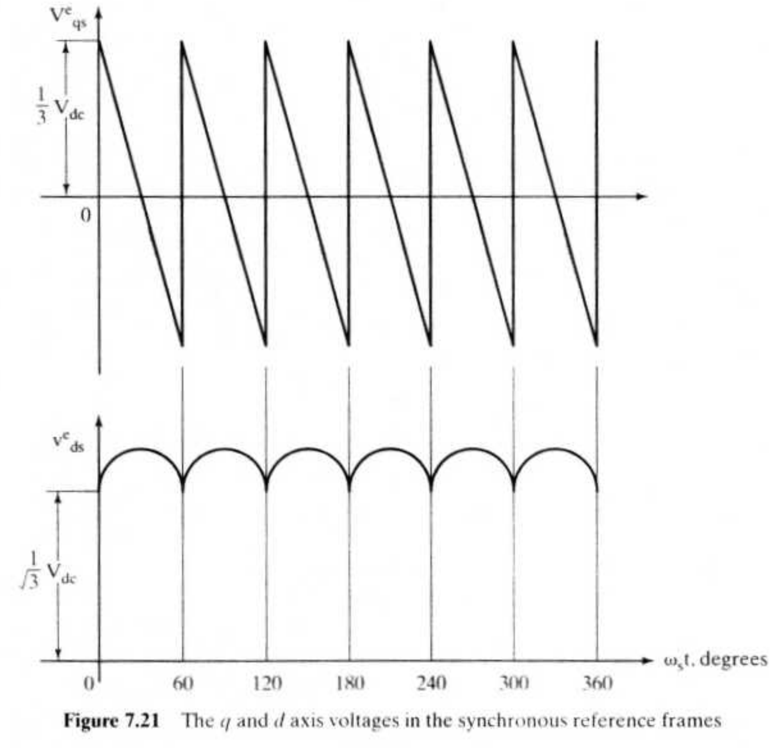

Figure 7.21 The q and d axis voltages in the synchronous reference frames